

LAB MANUAL: ANALYTICAL CHEMISTRY AT DUKE



CHEM 401L Analytical Chemistry

Duke University

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Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Orientation to CHEM401L

Learning Objectives

By the end of this module, you'll be able to:

1. Access all course details in the syllabus, Canvas site, and this manual. This includes:
 - Finding instructions and resources for experimental modules.
 - Understanding guidelines and due dates for written assignments.
 - Accessing rubrics for every assignment.
 - Knowing where to find course policies and support.
2. Explain the specific policies for CHEM 401L.
3. Effectively plan and prepare for lab meetings and stay on top of course deadlines.
4. Submit assignments through Canvas.

 **Purpose:** The Orientation Module introduces you to the structure, expectations, and tools used in CHEM 401L. Before beginning experimental work, it's important that you understand how this course is organized, how you will communicate your findings, and what is expected of you as a member of a collaborative laboratory team. This module will prepare you to navigate the semester successfully.

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1.1: Pre-Semester Assignment

 **Due:** All items in Part I below must be completed before the first day of class (11:59 pm on the Sunday day prior to FDOC). Part II must be completed prior to the first Wet Lab meeting, and Part III should be completed as necessary.

 **Time:** Plan at least three hours in the weeks prior to FDOC to complete Part I of this assignment. Parts II and III will take additional time.

 **Purpose:** These pre-semester assignments are designed to help you start the semester prepared and confident. You will get familiar with the course structure, Canvas site, expectations, and resources.

✓ Pre-Semester Assignment

Part I. Get Familiar with Course Tools and Policies

Complete the following written and preparation tasks before our first meeting:

1. Explore the Canvas Site
 - Log into Duke's Canvas platform.
 - Locate the CHEM 401L course site.
 - Carefully read the Course Syllabus (posted on Canvas).
 - Browse the Modules, Assignments, and Announcements sections.
2. Read the Orientation Materials on LibreText
 - Read the full "[1: Orientation to CHEM401L](#)" Module in the Lab Manual (this module).
 - Review the [Duke Chemistry Safety Manual](#) (linked in Canvas or the Lab Manual).
3. Complete the Orientation Assignment
 - Access and download the "Orientation Assignment" (at the end of the syllabus) from Canvas.
 - Complete the assignment carefully, following all instructions.
 - Submit your finished version as a PDF on Canvas before the first Dry Lab.
4. Download the full lab manual PDF in case of Canvas or LibreTexts outages.

Part II. Prepare for the First Week of Lab

1. Be Ready for Dry Lab (Monday)
 - Complete Part I above; Be prepared with any questions you may have about the Orientation materials.
 - Bring your computer to the first meeting.
2. Be Ready for Wet Lab (Sectional)
 - Complete the Pre-Lab Assignment for [Week 1, Module 2](#).
 - Know your assigned Wet Lab section and show up on time.
 - Dress appropriately: closed-toe shoes, clothes that cover your entire body from knees to elbows, your own eye protection.
 - Bring a lab notebook: you may use a physical notebook or electronic format (no template provided; use your training from CHEM 301L/310L).

Part III. Review Prerequisite Knowledge

You are expected to begin the course with working knowledge from prior chemistry and lab courses, especially:

1. General, Organic, and Physical Chemistry:
 - General chemical principles, units, and stoichiometry
 - Organic functional group reactivity
 - Thermodynamics, kinetics, and equilibrium
2. Physical Chemistry Laboratory ([CHEM301L](#) or [CHEM310L](#)):
 - Error Analysis & Reporting:
 - Types of experimental error
 - Propagation of error
 - Basic statistical analysis

- Reporting significant figures and uncertainty
 - How to write a report in the format of a scientific manuscript
- 3. Electronic Absorption Spectroscopy:
 - Beer-Lambert Law and calibration curves
 - Instrumental considerations (light source, detector, cuvettes)
 - Basic UV-vis theory

If you need to brush up on any of these areas, please revisit your prior course notes or request resources from the instructor.

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1.2: Welcome and Course Overview

Welcome to CHEM 401L!

This Advanced Analytical Chemistry Laboratory course is designed to simulate a research-intensive analytical chemistry experience. You will engage in authentic lab investigations, generate real data, and develop scientific arguments through writing. Success in this course requires engagement, reflection, and collaboration. Your instructors are excited to guide you through a semester of growth in both your technical and scientific communication skills.

Course Structure

The semester is divided into six parts:

1. Orientation Module (Pre-semester)
2. Calibration & Quantitation (2 weeks)
3. Project 1 (3 weeks), plus Week 4 for report revision
4. Project 2 (3 weeks)
5. Project 3 (3 weeks)
6. Final Reflection & Survey (Week 14)

Each week of the semester includes two meetings:

- Dry Lab (Mondays): Guided discussion, planning, and pre-lab work
- Wet Lab (Sectional): Hands-on experimentation, data collection, and data analysis

Assignments and Grading

The course totals 524 points from pre-labs, notebook submissions, reports, peer review, lab citizenship, and feedback surveys. A detailed breakdown of points in each assignment is available in this [google spreadsheet \(Click\)](#).

Assignments each week | each module:

- **Weekly Pre-Lab:** Pre-labs are due before each Monday Dry Lab.
- **Weekly Group Work:** At the end of each Dry Lab, your group will submit a group discussion report.
- **Weekly Notebook:** At the end of each Wet Lab, you'll submit a notebook snapshot with a brief informal reflection.
- **Each Module:** At the end of each Module, you will submit a formal report.
- **Each Module:** At the end of each Module, you will complete a survey to provide feedback to your instructors about project design and learning support.

Promoting Diversity, Inclusion, and Avoiding Harassment



All members of this course will foster an environment of listening and considering new ideas from a diverse group, with respect for all participants without regard to gender, race, ethnicity, color, sexual orientation, political orientation, age, religion, national origin, disability, or any other aspect of how we identify ourselves other than as fellow scientists.

There is zero tolerance in this course for any form of harassment or hate speech, during class meetings or outside of scheduled class time, which could include verbal or physical conduct that has the purpose or effect of interfering with anyone else's

participation or performance, or of creating an intimidating, hostile, or offensive environment; any such harassment will result in dismissal from the course and will be reported to the college.

Duke University's Institutional Statement of Commitment to Diversity and Inclusion is found [here \(click\)](#):

"Duke aspires to create a community built on collaboration, innovation, creativity, and belonging. Our collective success depends on the robust exchange of ideas—an exchange that is best when the rich diversity of our perspectives, backgrounds, and experiences flourishes. To achieve this exchange, it is essential that all members of the community feel secure and welcome, that the contributions of all individuals are respected, and that all voices are heard. All members of our community have a responsibility to uphold these values."

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1.3: Details about Meetings and Deadlines

Where to find Information about this course

1. **Canvas:** Our course site is on Canvas. All course information is accessible through links on Canvas. All assignments will be submitted electronically on Canvas, and all rubrics are available in the assignments on Canvas.
2. **Lab Manual:** The lab manual lives on LibreTexts, and this is where you find daily tasks and agendas. You can download a copy of the lab manual at any time. Please note that the manual is a "living" document that we update regularly. The latest version is available on LibreTexts. We recommend that you download a pdf copy as a backup for times when you need to work offline. The Lab Manual will NOT be available in Wet Lab - you must have all procedures and plans written in your notebook ahead of time.
3. **Syllabus:** The syllabus contains additional details and policies that are not provided in the public manuals on LibreTexts, including instructor contact information. The detailed schedule of deadlines and assignments is also in the Syllabus document.

Deadlines

Submit your assignments and arrive to meetings on time to receive the maximum points for your work. See the schedule (syllabus) for a detailed schedule of all deadlines.

General Assignment Deadlines

- **Pre-Lab assignments:** these are due electronically by 10:00 am on the day of the "Dry Lab" meeting (Mondays).
- **Arrival to Lab:** Be ready to begin at the start time of each meeting. Arriving 5 minutes early is recommended.
- **Dry Lab Group Work:** The reporter from each group will submit a record of the group's work at the end of each dry lab session.
- **Notebook and Reflection:** A snapshot of your day's notebook and reflection is due at the end of each wet lab meeting.
- **Reports:** Report drafts are due at the third dry lab of each module; peer reviews are due by the end of that dry lab meeting. Final reports are due by the subsequent dry lab meeting. (see syllabus schedule)

Extensions and Late Assignments

This course is designed to keep you on track while also recognizing that real life happens. The policies below aim to balance accountability with flexibility, providing you reasonable options when things don't go as planned.

Policy on Extensions and Late Assignments

General Policy

- Deadlines are essential to ensure that you are prepared for lab work, receive timely feedback, and stay aligned with your team and the instructional schedule.
- Accommodations are always available for students facing serious or ongoing extenuating circumstances. If this applies to you, contact your academic dean to request a formal Dean's Excuse; your Dean will then reach out to your instructors so we can support you appropriately.
- The policies below apply to all students in the absence of a Dean's Excuse.

Late Assignments: If an assignment is not turned in by the deadline, the following point deductions will apply.

- **For Pre-Lab assignments:** Pre-labs are required preparation for Dry Lab and Wet Lab sessions. The policy for pre-labs is the most strict to protect your safety, the safety of expensive equipment, your team's success, and your ability to contribute meaningfully to lab work.
 - Pre-labs must be submitted before the Dry Lab session begins.
 - Late pre-labs will receive zero points.
 - If the pre-lab is still incomplete before your scheduled Wet Lab, you will not be allowed to participate in the experiment.
- **For All other assignments:** All other assignments can be submitted late but incur a deduction.

- 10% deduction per 24-hour period, up to a maximum of 50% off.

Examples:

- 5 minutes late → 10% off
- 26 hours late → 20% off
- More than 5 days late → 50% off

- **Deadline for All Late work:** Late work is only accepted up to 11:59 pm on LDOC (Last Day of Classes).

Extension Token: Each student may use one Extension Token during the semester — no explanation required.

- **The token allows you to submit:**

- A scientific report up to two weeks late with no penalty, OR
- Any other assignment up to two weeks late, as long as it doesn't impact safety, team grading, or lab scheduling.

- **⚠ Use It Wisely:**

- This token is not a substitute for a Dean's Excuse. If you're dealing with a serious situation, ask your Dean or your instructional team for a different level of support.
- Using the token may delay feedback or grading, which can impact your ability to perform on subsequent assignments. Do not use this as a license to procrastinate!.

Final Deadline for Late Work

No late work will be accepted after 11:59 PM on the Last Day of Classes (LDOC).

Preparing for and Participating in Lab Meetings

This section outlines expectations for your preparation, participation, and behavior in both Dry Lab and Wet Lab sessions. These practices support your safety, efficiency, and success in a collaborative research-focused environment.

Before Lab: Preparation is Critical

- **Complete the Pre-Lab Assignment and submit it on Canvas BEFORE Dry Lab each week:** It is essential that each student be prepared for lab meetings by reading the appropriate module in this manual, and completing the specified pre-laboratory assignments. The pre-laboratory assignments are designed to prepare you to learn, operate efficiently, protect our equipment, and keep you safe during the experimental module. Each pre-lab assignment requires you read the entire module and write out the specified pre-lab assignments in your notebook. Be sure to have a work plan that makes efficient use of your time, and take notes that will help you during wet lab. Failure to do this will undoubtedly result in your failing to complete the laboratory in the allotted time (see "leaving the laboratory"), and therefore decrease the chance that you will earn desirable scores on your assignments.
 - The Pre-Lab is due before your Dry Lab; late pre-labs will receive a score of zero.
 - You will not be allowed to conduct experiments if the pre-lab assignment is not completed before your Wet Lab meeting.
 - If there is an extenuating circumstance that prevents your preparation, please inform your instructors as soon as possible.
- **Dress for Lab:** Dress appropriately for a laboratory environment. You must wear close-toes shoes and clothing must cover your body from your upper arms to your knees, including your torso. You should also have your own pair of protective eyewear, bring them to lab, and wear them the entire time you are in lab. [Safety eye wear can be purchased from the Graduate Chemistry Council \(GCC\)](#). If lab attire is not jiving with your wardrobe, keep a change of clothes on hand or purchase a lab coat and bring it with you so that you can cover up in lab. Students who are not dressed appropriately will be dismissed and not allow to complete the experiment.

During Lab: Expectations for Wet Lab Sessions

- **Stay on task during the lab period:** Each lab module is designed so that a well-prepared student can complete the laboratory work within the scheduled laboratory meeting times. You will complete your work individually, and in some cases, you will share your results with your group members and analyze the data as a group. There will a dedicated TA to help you run the instruments, and answer questions. Lengthy sample preparation procedures should be started early in the lab period.
- **Record your work:** All students are expected to record laboratory observations and any numerical data in a dedicated laboratory notebook for this course; the notebook may be a physical bound notebook or an electronic notebook (eg OneNote or LabArchives). Each page should be titled, dated, and signed. You should have a way of collecting and organizing any

instrument outputs (data), such as spectra and chromatograms. Every lab computer has internet access, and you may either email data to yourself or collect it using a usb drive. A snapshot of your notebook must be submitted at the end of each Wet Lab session.

- **Lab Etiquette:** During the lab period, you should actively maintain the shared spaces by cleaning up after yourself. Clean up spills immediately, brush powders off of the balances after every use, and put all trash or waste in the appropriate containers. At the end of each lab, you should leave the lab space, equipment, and instruments in good condition. Clean all glassware, remove any labels from the glassware, and return it to the appropriate location. Wipe down and clean the surfaces and equipment that you used and leave the lab better than as you found it.
- **Use the time with your instructors (and bring snacks to avoid hanger!):** The lab period is loooong for a reason! A big part of your work for this course includes data analysis and writing. There will be several occasions when your laboratory work is finished before the lab period ends. This "extra" time is deliberate - you should use this time to get your data off of the lab computers and discuss it with your instructors or your group. During scheduled lab hours is the best time to get help from your instructors! Do not squander the downtime during scheduled lab meetings! We recommend that you bring a snack to eat (Eat in the hallway or in conference room 1113)! A quick snack can help you have the energy to use the end of lab hours most effectively!
- **Leaving Lab:** In the interest of safety, we do not allow students in the lab after business hours (ie you do not have access to lab between ~ 6 pm - 9 am). Thus, you must leave the lab promptly at the end of the period. At 10 minutes prior to the end of the period, you must stop lab work and begin cleaning up (see lab etiquette). Your instructors must sign you out of lab before you leave for the day; they will record the time you leave (be sure they do this!) after checking that your space is tidy and all materials have been returned to the appropriate places.

Wet Lab Safety

The educational objectives of this course include learning about the hazards of materials, equipment and procedures in the chemical laboratory. This is very important because it determines the level of risk to which you and other students are exposed by your actions and the actions of others while you are in the laboratory. Through this type of training, you will gain the knowledge and experience to determine what precautions are necessary in order to limit risk to yourself and others from chemical hazards. We hope that you will make a serious effort at all times to ensure that your work in the laboratory is conducted as safely as possible. The following rules have been developed with your safety in mind:

- **SAFETY GLASSES MUST BE WORN AT ALL TIMES IN THE LABORATORY-NO EXCEPTIONS.** Contact lenses may be worn in the laboratory as long as you are also wearing safety glasses. You are required to bring your own safety eye wear. [Safety eye wear can be purchased on campus from the Graduate Chemistry Council \(GCC\).](#)
- A copy of the [Duke Chemistry Department Safety Manual](#) is available on the Chemistry web site. All students should read this manual prior to beginning laboratory work.
- No unauthorized experimentation will be allowed in the laboratory.
- Proper shoes are required in the laboratory - no bare feet, sandals or open-toe footwear.
- Never eat (including chewing gum), drink, or smoke in the laboratory.
- To minimize skin exposure to chemical hazards, skin must be covered from your shoulders and upper arms to your knees. Sleeved shirt and pants should be worn. We know it can get hot in Durham, so bring extra clothes with you to lab on hot days. Clothing that are not appropriate for lab include shorts, mini-skirts, crop tops, tank tops, low cut shirts, etc.
- Long hair should always be tied back while in the laboratory.
- Safety Data Sheets (SDS) and other safety notes for every chemical to be used in this laboratory course are kept in a ring binder in the lab and also online. Students should read these notes prior to beginning each experiment.
- Failure to observe safety regulations will result in substantial deduction in the lab grade and may result in ejection from the lab session. Persistent violation of safety regulations will result in permanent expulsion from the lab course.
- **Disposal of Solvents and Chemicals: Never** pour solvents or chemicals down a drain. Departmental waste disposal procedures are described in the safety manual. However, you should **ALWAYS** consult with a TA before attempting to dispose of any material.
- Precautions for individual experiments will be discussed with you by the TA's and are described in the lab manual.

Working with Your Instructional Team

- **Your instructors are here to help:** There is a team of instructors to help you, including the course instructor (a Ph.D. chemist) and Teaching Assistants (TA's) who are graduate students earning their PhD's in chemistry. There will be TA's in the lab with

you each day, and at least one of them will be dedicated to helping you on each experiment. That dedicated TA will be present during the laboratory to assist you, offer tips, and advice. That same TA will be on the team of TAs that grades your lab report for that given experiment. A list of instructors and their contact information is posted on the syllabus.

- **Use lab time to get help:** You are strongly advised to consult with your TA about a given report before writing it. This consultation should happen during scheduled lab time when possible - do not leave early just because you have finished the experiment and cleaned up. If you complete the experiment in the first lab meeting, use the second lab meeting as a consultation with your TA. Use lab time wisely to work on your data and consult with your TA so they can help you during your time with them. The TA can show you the important points to be emphasized in evaluating your data and will discuss the grading scheme.
- **Email:** You may also email your TA at least 48 hours before your lab report is due to ask questions about an assignment; do NOT wait till the last moment to do this because you will not get help in time. Please be polite in your email correspondences. Please also recognize that questions about chemistry are often difficult to answer in an email, and a meeting may be much more effective. Please take advantage of lab time and office hours for this reason.
- **Office Hours:** The office hours schedule with your team of instructors is posted on the Syllabus.

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1.4: Academic Integrity

Academic Integrity vs Cheating

- **Academic integrity** means your work is an accurate representation of your own learning and understanding.
- **Cheating** or **academic dishonesty** occurs when your work misrepresents what you know, undermining your own learning and the trust that makes meaningful feedback possible.

Why Academic Integrity Matters in this Course

In this course, your written work is both a learning tool and an assessment tool.

- When you write, you articulate and refine your thinking.
- When we read your writing, your instructors and TAs see your reasoning, your decisions, and your conclusions. This allows us to give you feedback you can actually use to grow as a scientist.

You will also give and receive peer feedback. For that feedback to be meaningful — and for you to benefit from it — your work must reflect your own thinking.

Duke Community Standard

All students must adhere to the [Duke Community Standard \(DCS\)](#): Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, and accountability. Citizens of this community commit to reflect upon these principles in all academic and non-academic endeavors, and to protect and promote a culture of integrity.

To uphold the Duke Community Standard, students agree:

- will not lie, cheat, or steal in my academic endeavors;
- I will conduct myself honorably in all my endeavors;
- I will act if the Standard is compromised.

Regardless of course delivery format, it is the responsibility of all students to understand and follow all Duke policies, including academic integrity (e.g., completing one's own work, following proper citation of sources, adhering to guidance around group work projects, and more). Ignoring these requirements is a violation of the Duke Community Standard. Any questions and/or concerns regarding academic integrity can be directed to the Office of Student Conduct and Community Standards atconduct@duke.edu.

Using AI to Support Your Learning

AI tools (e.g., ChatGPT, Grammarly, Claude) can be powerful learning partners if you use them intentionally and transparently. But, if they are misused, these tools can undermine your learning.

Appropriate use of AI:

AI tools can play a valuable role in your learning process when used intentionally to coach you, or to help improve your writing.

You may use AI applications to:

- Improve organization and structure of your own writing using prompts like *“Revise this paragraph to improve clarity and structure, but keep the original meaning.”*
- Edit for grammar and clarity using prompts like *“Identify grammar errors and suggest fixes without changing the technical meaning of my writing.”*
- Refine your phrasing or writing style. *“Make this explanation appropriate for an audience of scientists outside my specialty.”*
- Brainstorming ways to present an idea you’ve already developed. *“Suggest alternative ways to phrase this description of my method.”*

After using AI in any of these ways, you must review, edit, and verify the output yourself. Be aware that AI tools can change the meaning of your words, and AI can hallucinate and give test that is not based in reality. You are responsible for ensuring that your final writing is clear, correct, and accurate — and that it represents your own learning, thoughts, and understanding.

Inappropriate use of AI:

Using AI to generate substantive scientific content — such as designing methods, interpreting data, or drafting analysis you have not developed or do not fully understand — is a misuse of these tools.

Examples of inappropriate use of AI:

- Generating new sections or paragraphs you did not develop yourself. (e.g., “Write an introduction section for a scientific manuscript about ...”)
- Asking AI to analyze your data or draw conclusions that you cannot explain (e.g., “Interpret my results and write the discussion section.”)
- Rewriting entire drafts you did not author yourself.
- Fabricating sources, citations, or data.

Misusing AI in this way prevents your instructors and peers from providing meaningful feedback; it prevents you from learning and it wastes everyone's time.

A Quick Test: The “Could I Explain This?” Rule

Before submitting any work that involved AI, ask yourself:

- Could I explain every choice, result, and statement in my own words?
- Could I recreate the reasoning or analysis without AI's help?

If the answer is “no,” then the AI use is likely inappropriate for this course.

When in Doubt, ask.

If you are ever unsure about whether an AI tool is supporting your learning appropriately, please ask a member of the instructional team. We are here to help you learn how to use these tools thoughtfully, responsibly, and always in service of your authentic growth as a scientist and writer.

Acknowledgements

Whenever you receive outside help—whether from a friend, a textbook, or an AI tool—you must acknowledge it. To help your instructors understand what kinds of support you are using and where you may need additional guidance, we ask that you clearly document all sources of assistance for each assignment.

Example of an Acknowledgements statement:

This statement on the responsible use of AI tools was created by Kathryn Haas for Advanced Lab Courses at Duke University. The statement was drafted with the support of OpenAI's ChatGPT. It draws inspiration from “AI with Intention: Principles and Action Steps for Teachers and School Leaders” by Tony Frontier and includes insights generously shared by Dr. Eliana Schonberg (Thompson Writing Program at Duke University).

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1.5: Scientific Writing and Report Grading

Why Scientific Writing... in a Chemistry Lab Course?

Writing is equivalent to thinking. By writing about your experiments, you deepen your understanding of what you did and why it matters. Scientific writing is not just a course requirement — it's a central part of doing science. In this course, you will complete three "scientific-manuscript-style" reports using the skills and formats you learned in CHEM301L or 311L. Refer to the [Writing Support Materials](#) from CHEM301L for support.

Support for Writing

This course is designated as a [Writing \(W\)](#) course at Duke, which means that developing your scientific writing skills is a central learning goal. Dr. Haas has been teaching writing-intensive courses since 2014 and has worked closely with the Thompson Writing Program and the graduate TAs to ensure the entire instructional team can provide strong, consistent writing support.

In addition to feedback from your instructor and TAs, you will have access to the [Thompson Writing Studio](#) and an Embedded [Writing Consultant](#) (EWC) assigned to this course. EWCs are trained undergraduate peer tutors who will work with you to strengthen your writing process and help you communicate your science effectively.

You will meet with your EWC during the third Wet Lab session of each module, giving you the opportunity to get targeted feedback on your work in progress. The [Thompson Writing Studio](#) is also available for one-on-one appointments at any stage of your writing process.

Report Expectations

All reports (except for Module 2) must follow the structure of a scientific manuscript: Title, Introduction, Methods, Results, Discussion, Conclusions, and References. Use the formatting conventions you learned in prior lab courses (Refer to the [Writing Support Materials](#) from CHEM301L for support). Your goal is to present your analysis clearly, concisely, and accurately — not to write a perfect or lengthy document.

Submission Guidelines

Submit your report as one single pdf file on Canvas. Your report will be graded by a group of TA's. When TA's grade your report, they will not see your name (except on the title page) - this is to reduce any possible biases during point allotment.

Grading Criteria

Reports are scored using detailed rubrics developed by the instructional team. These are available in each Canvas assignment. A Generic version of the Rubric for a Report in the format of a scientific manuscript is given below. This generic rubric was developed with support from Dr. Shonberg of the Thompson Writing Program at Duke.

Component	Points
Title / Author	5
Introduction	15
Experimental Methods	10
Results	20
Discussion	20
Conclusion (as part of Discussion)	5
References	10
Writing Quality	15
Total	100

Writing Quality Policy ("B or Nothing")

Each section of your report must be B-quality ($\geq 80\%$) or better to earn points. If a section does not meet that standard, it receives zero points.

To support learning:

- Each Pre-lab assignment is designed to make progress, and for you to receive early feedback, on parts of your report.
- Embedded Writing Consultants will meet with you to give you individual advice on your writing.

- You may revise and resubmit the first report after receiving feedback from peers and from your instructors.
- Each report will have an early deadline, one round of peer review, followed by time for revisions, and a final submission at least one week later.
- You may submit one report late without penalty using your Extension Token.

Tips for Success in Report Writing

- Every assignment in this course — pre-labs, notebook entries, data analysis tasks — is designed to help you build your report piece by piece. Treat them as stepping stones toward your final manuscript.
- Use your time strategically:
 - Start drafting sections of your report during or right after lab, while your methods, observations, and insights are still fresh.
 - Create figures and plots as soon as you have your data, and review them with your TA to check for clarity and accuracy.
 - Use lab time for discussion—ask questions, clarify your interpretation, and get feedback before you start formal writing.
 - The sooner you start, the easier it will be to produce a clear, well-organized report without last-minute stress.

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- [Introductory Details](#) by W. R. Fawcett, John Berg, P. B. Kelley, Carlito B. Lebrilla, Gang-yu Liu, Delmar Larsen, Paul Hrvatin, David Goodin, and Brooke McMahon is licensed [CC BY-NC-SA 4.0](#).

1.6: Lab Performance and Professionalism

A portion of the total available points for this course are allotted to your performance in lab. This includes an expectation for preparation before the lab period, arriving on time, using time effectively, performing the tasks well, contributing to the culture of safety, and helping to maintain a clean and safe environment. Your performance and professionalism in the lab is assessed through **Lab Citizenship points**.

What does it mean to be a "Good Lab Citizen"?

Lab Citizenship refers to more than just following rules. It means self-regulating your behavior in a way that supports a safe, respectful, and productive lab environment—without needing constant reminders from instructors.

Your instructors are here to support your learning, not to act as the "lab police." We expect that you will take ownership of your space, your preparation, and your conduct. This includes not just compliance with safety policies, but also contributing positively to the shared environment, being considerate of others, and acting professionally.

Good Lab citizens:

- Arrive on time and ready to work
- Wear proper lab attire and PPE at all times without reminders
- Turn notifications off on electronic devices used during meetings
- Clean up after themselves and leave shared spaces better than they found them
- Use lab time effectively and help their group stay on track
- Label solutions, handle chemicals safely, and follow disposal protocols
- Respect the physical space, equipment, and people in the lab

When you do these things consistently, it frees your instructors to focus on helping you learn and grow as scientists — not on enforcing rules. Your Lab Citizenship grade score reflects your ability to self-regulate and contribute to a positive lab culture.

Lab Citizenship Checklist

Each Wet Lab session, you can earn 1 Lab Citizenship point by demonstrating professional, safe, and respectful lab conduct. This point is awarded as an "all-or-nothing" score based on the following expectations.

Daily Lab Citizenship Expectations (1 pt)
✓ Arrived on time and stayed the full session
✓ Worked safely and efficiently; used time well*
✓ Wore appropriate PPE and clothing without reminders
✓ Cleaned glassware, workspace, and instruments used
✓ Labeled all solutions and disposed of waste properly
✓ Maintained a respectful, team-oriented lab environment

 Students **will not receive partial credit**—failure to meet any of these expectations may result in 0 points for the day.

*Distraction by electronic devices (messages, social media) during lab time is considered a violation of safety and using time well. You are advised to turn off notifications on electronic devices used during meetings.

Lab Citizenship Penalties

While Lab Citizenship is typically graded on a 1-point-per-session basis, certain behaviors may result in point deductions from your lab report score. These behaviors compromise safety, professionalism, or lab functionality and require stronger consequences.

Repeated or serious infractions will result in direct point deductions from your lab report grade for the corresponding module. The instructional team will notify you if a deduction has been applied and explain why. Some, but not all, examples are shown below:

Infraction	Deduction from Report Score
7—15 minutes late to lab	–5 points
16—20 minutes late to lab	–10 points
More than 20 minutes late	–15 points (not allowed to participate)
Failure to wear safety goggles (after first warning)	–5 points per occurrence
Improper lab clothing (after first warning)	–10 points per occurrence
Working while visibly ill without a face mask	–5 points per occurrence

! Repeated issues (e.g. forgetting goggles, being late, or poor cleanup) may also trigger **a conversation with the instructional team or department administrators** to resolve concerns and may result in failure of the course.

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CHAPTER OVERVIEW

2: Calibration and Quantitative Techniques

Learning Objectives

By the end of this module, you will be able to:

1. **Design and prepare a set of standard solutions** using appropriate volumetric techniques to support a quantitative analysis.
2. **Construct and interpret a calibration curve** using spectrophotometric data collected at a specific wavelength.
3. **Evaluate the quality of calibration data** using residuals and statistical evidence to justify regression model choice.
4. Apply a calibration model to **calculate the concentration of an unknown sample**, including propagation of error.
5. **Document experimental work clearly** and completely in a scientific notebook.
6. **Prepare publication-quality figures with captions.**
7. **Engage in safe and responsible laboratory practices**, demonstrating awareness of good lab citizenship.

The purpose of this introductory lab module is to practice using some of the basic quantitative techniques that you will use often in this course and in an analytical chemistry laboratory. These techniques include the use of an analytical balance, quantitative use of pipettes and volumetric glassware, quantitative solution preparation and dilution procedures. Although you may have used these basic equipment in previous laboratory courses, **analytical chemistry requires a *much higher* level of precision and accuracy than you may have experienced in general or organic chemistry courses.** One of the requirements for success in this course is the development of correct quantitative laboratory technique.

[2.1: Introduction](#)

[2.2: Week 1](#)

[2.3: Week 2](#)

[2.4: Post-Lab](#)

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2.1: Introduction

Introduction

In this two-session lab module, you will take on the role of a method developer. Your task is to evaluate how different calibration design choices influence the quality of analytical data. You will construct and compare calibration curves using visible-range absorbance measurements of a colored dye, bromocresol green.

Rather than simply following instructions, your goal is to generate, validate, and evaluate calibration curves using different tools and dilution strategies. You will also share your data with the class to support broader comparative analysis.

Bromocresol green

Bromocresol green ($pK_a = 4.66$) is an acid-base indicator dye that exhibits a pH-dependent absorption spectrum. The only structural difference between the **acid form (yellow)** and the **basic form (blue)** of this dye is the protonation of a phenolic oxygen. This experiment will be performed in 0.01 M sodium acetate; this weak base in solution will ensure that the pH is close to 8 and that bromocresol green dye is predominantly in its **basic form**.

A stock solution of bromocresol green will be provided to you in lab. You will use this *stock solution* to prepare a series of more dilute solutions. You will measure the absorbance of the diluted solutions at **612 nm (absorption maximum for the basic form of the dye)** and generate a *calibration curve* of absorbance vs concentration. You will then measure the absorbance of a bromocresol green solution of unknown concentration, and use Beer's law to determine its concentration.



Figure 2.1.1: Top: acid dissociation of bromocresol green. Bottom, from left to right: five solutions of 0.1 M HCl with decreasing concentrations of bromocresol green (yellow); three buffered solutions at pH 3.78, pH 4.00, and pH 4.62 (shades of green) and four solutions of NaOH 0.1 M with increasing concentrations of bromocresol green (darker solutions are more concentrated). ([Natan Consigli](#), [Bromocresolo](#), crop, [CC BY-SA 4.0](#))

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2.2: Week 1

Pre-Lab Assignment

Due: This pre-lab assignment is due prior to the first Wet Lab meeting, on the first week of classes.*

*The pre-lab assignments in this course are due prior to arriving at the Dry-Lab meetings — *except for this first module*. Due to the unique schedule in the first two weeks of the semester, the first two pre-labs will be due prior to arriving at Wet Lab meetings. Please note that starting the third week, all pre-lab assignments are due prior to the dry lab meeting (Monday)!

Purpose: In this module, you will plan, record, and analyze calibration data - this module is practice to get you warmed up for the three projects later in the semester. This pre-lab assignment will help you come prepared to measure solutions carefully and think about precision and accuracy.

Week 1 Pre-Lab Tasks

1. Read & Review

- Read Module 2 (this module) in the Lab Manual.
- Find the [Instrument Facilities Manual](#)
 - Read about how to use the [Analytical Equipment](#) (and watch the videos) - note the Tolerances (errors) of common equipment at the end of the page!
 - Read about [Solution Preparation](#) and practice the Error Propagation exercises.
 - Read about [Methods of Calibration](#): As you read, think about which type of calibration you will use in this module.*
*This is for your own reflection—no written response is required for submission or grading.
 - Read about how to [Determine uncertainty from values determined from a Calibration Curve](#). Since we did not use calibration in P-chem lab, this may be completely new to you. Expect to see this new material in CHEM401L lecture!

2. Prepare your Notebook for Week 1 (Do the following in your notebook and turn it in on Canvas)

- Take notes** on the tools you'll use, solutions you'll make, and measurements you'll take.
 - Summarize the procedure in your own words.
 - Summarize the main points for proper use of volumetric pipets (with one t) and pipettes (with two t's).
 - Record the tolerances for all glassware and pipettes you'll use in this module!
- Get to know your analyte (Bromocresol Green)**
 - Draw the chemical structure of bromocresol green in its acidic and basic forms.
 - Note which form you will measure and the wavelength you'll use.
- Predict what you'll see & plan what you'll do**
 - Sketch a predicted calibration curve from solutions B-F (Absorbance vs. Concentration).
 - Label the standards (B-F), the blank, and the validation standard on your plot.
 - Show how you will calculate the concentration for at least one standard using $M_1V_1 = M_2V_2$.
- Prepare Data Tables in your notebook (see examples below).** Build a data table with the following:
 - Rows for standards B-F, control, unknown
 - Columns for 3 replicate absorbances, average, standard deviation

Example Notebook Tables

Prepare tables in your notebook for recording data in an organized way (see examples below). In the same table, provide space for summarizing propagated error in concentrations and the standard deviation of replicate absorbance measurements of each solution.

If you choose to record your data electronically, please use a shared document that your team can access and so you can collaborate on a shared file. Also, be sure to record a link or the file path to the electronic files in your notebook. There are example data tables for this module available in this [google sheet](#) (click). A screen shot of the example data tables are below.

Example table for Calibration 1, Trial 1:

Volumetric pipet Trial 1 data				Replicate measurements of absorbance (AU)				
Solution Label	Components*	Actual Concentration of Bromocresol	±	Propagated error	1	2	3	Average Absorbance ± Standard deviation
Stock solution A	Stock solution* of Bromocresol Green (0.500 mM)	5.00E-04	±		NA	NA	NA	NA ± NA
Calibration Standard B	1.000 mL of stock A, diluted* to 25.00 mL	2.00E-05	±					±
Calibration Standard C1	15.00 mL of B, diluted* to 25.00 mL	1.20E-05	±					±
Calibration Standard D1	15.00 mL of C, diluted* to 25.00 mL	7.20E-06	±					±
Calibration Standard E1	15.00 mL of D, diluted* to 25.00 mL	4.32E-06	±					±
Calibration Standard F1	15.00 mL of E, diluted* to 25.00 mL	2.59E-06	±					±
Blank	0.01 M sodium acetate	0	±					±
Validation Standard Sample			±					±

*All samples are in a matrix of (and diluted with) 0.01 M Sodium acetate solution.

Example table for Calibration 2, Trial 1:

Micropipette Trial 1 data				Replicate measurements of absorbance (AU)				
Solution Label	Components*	Actual Concentration of Bromocresol	±	Propagated error	1	2	3	Average Absorbance ± Standard deviation
Stock solution A	Stock solution* of Bromocresol Green (0.500 mM)	5.00E-04	±		NA	NA	NA	NA ± NA
Calibration Standard B	1.000 mL of stock A, diluted* to 25.00 mL	2.00E-05	±					±
Calibration Standard C2	1.80 mL of B, diluted* to 3.00 mL	1.20E-05	±					±
Calibration Standard D2	1.80 mL of C, diluted* to 3.00 mL	7.20E-06	±					±
Calibration Standard E2	1.80 mL of D, diluted* to 3.00 mL	4.32E-06	±					±
Calibration Standard F2	1.80 mL of E, diluted* to 3.00 mL	2.59E-06	±					±
Blank	0.01 M sodium acetate	0	±					±
Validation Standard Sample			±					±

*All samples are in a matrix of (and diluted with) 0.01 M Sodium acetate solution.

Dry Lab Agenda

This Dry Lab meeting is the first meeting of the semester.

1. **Course Welcome:** Brief course introduction and Q&A.
2. **Safety talk and Tour** of the Laboratory Space.
3. **Discussion about first Wet Lab session:** Come prepared with questions from your Pre-Lab.

Wet Lab, Week 1: Comparing Tools (Volumetric vs. Automatic pipettes)

 **Materials:** Gather the following and inspect all equipment prior to use.

1. Bromocresol green solutions:

- o **Solution "A",** a **"stock solution"** of 5.00×10^{-4} M bromocresol green in 0.1M sodium acetate (prepared by the instructional team by combining 0.0900 g of the sodium salt of bromocresol green (MW 720.02 g/mol) and 0.3402 g of sodium acetate trihydrate in a 250.00 mL volumetric flask, then dissolving in DI water and filling the flask to the t.c. volumetric mark). The *exact concentration should be written on the stock container*. Be sure to record the *actual concentration* in case it deviates from what is written in this manual.
- o **Validation Standard** solution (aka "control solution") of known concentration. This is a solution of bromocresol green in 0.1 M sodium acetate. You will use this solution to **validate** your standard curve. *The exact concentration should be written on the control solution storage container*. Be sure to record the concentration in your lab notebook.
- o **Sample** solution (a solution of bromocresol green of unknown concentration in a matrix of 0.1 M sodium acetate). You should use your validated calibration curve to determine the concentration of this sample.

2. Matrix solution:

- o 125 mL of **0.01 M sodium acetate:** Prepared by the instructional team in advance by dissolving 2.7216 g of sodium acetate trihydrate (MW 136.08 g/mol) in 2.00 L of deionized water.

3. Equipment:

each team will share the following equipment depending on availability.

- o One **1000 μ L automatic micropipette**
- o One **2.000 mL volumetric pipet** (optional for making solution B)
- o At least one **plastic tip** for the micropipette
- o One **25 mL volumetric flasks** (you could use several if available)
- o One **50 mL volumetric flask** (optional for making solution B)
- o Two disposable **transfer pipets** (to be reused throughout the solution preparation)
- o Glass vials with screw cap lids for storage of solutions
- o Two glass **cuvettes**
- o **Spec 20 instrument**
- o One **beaker** to collect waste

Important: Inspect your equipment

The function of the volumetric pipets and micropipettes depends on their condition! Carefully check their condition prior to use.

Volumetric (Class A glass) pipets:

- If the tip of the pipet is chipped, do not use it! Please alert your TA and get a new pipet.
- Visually inspect the pipet to be sure it appears clean.
- Rinse the pipet with your solution and drain it - visually inspect to ensure liquid is draining cleanly and there are no beads of solution on the inside of the pipette walls. If solution is beading up on the walls, either clean the pipette before use, or get a new one.

Micropipettes:

- Be sure that the plastic tip fits tightly onto the pipette barrel.
- Dial the pipette to 1000 microliters (**this looks like "100" on most of the p1000 pipettes**). Use a balance and a plastic weigh boat to test the pipette to be sure that it is delivering the correct volume. 1000 microliters = 1.000 mL = 1.000 grams. Consider the error expected when deciding whether to use the pipette.
- Repeat for 500 microliters.

Preparation of calibration standards

You will make a series of sequential or "serial" dilutions of bromocresol green and measure the absorbance of each solution in the series using a single-wavelength spectrometer. You will use these absorbance readings to construct a calibration curve and determine the concentration of an "unknown" solution. Each group (at least two people) should work together to create one validated calibration series using volumetric pipets, and one using automatic micropipettes, as described below. For groups of three and four teammates, do additional trials of each method depending on the number of additional people. Each person should produce at least one calibration set.

Note the tolerance of all equipment in your lab notebook so that you can propagate error in the concentration of each solution.

Calibration Standard "B"

You may collaborate and share this solution with your teammates or with other teams - however you are each individually responsible for the accuracy of this solution preparation and for recording how it was created in your notebook. **It is essential that the concentration of calibration standard B solution is accurate.**

- For convenience, use a micropipette and a volumetric flask: Use a **1.000 mL automatic micropipette** to quantitatively transfer 1.000 mL of stock solution to a **25.00 mL volumetric flask**. *It is critical that this transfer is precise and accurate — check the function of the pipette first, and [then weigh your transferred liquid](#) to be sure.* Then dilute with the matrix.
- When using the volumetric flasks, carefully dilute the solution up to the mark using **0.01 M sodium acetate solution** and mix well. Put some of this solution into a screw cap vial labeled "B" so that you can access the solution with a pipette in the next step. Each team will need at least ~20 mL of this solution or more. If you need to re-do the solution prep you will need even more. You can store the leftover solution B in a sealed vial.

- [Record triple replicate absorbance of Solution B](#) on at least two spectrometers (to be sure the spectrometers are giving consistent readings).

Calibration Standards C1-F1 — serial dilution, Class A volumetric glassware (Calibration Set 1)

One person in your team should create standard solutions for calibration set 1 (C1-F1) by serial dilution, using Class A volumetric glassware:

1. Start with solution B. Prepare solution "C1" using a 15 ml volumetric pipette, quantitatively transfer 15 ml of solution B to a clean 25 ml volumetric flask. Carefully dilute the solution up to the mark using 0.01 M sodium acetate solution. Mix well by inverting 20 times. Transfer solution C1 to a labeled vial before the next step.
2. In the same manner, perform 3 more serial dilutions to create solutions D1-F1 (see the example tables provided below).
 - Transfer and store each solution to a labeled vial and keep vials closed to avoid evaporation.
3. Collect absorbance data using the "Data Collection" instructions below. Make sure to also collect the absorbance of the Blank, the Validation Standard, and the Sample.
4. Use the data to determine if your calibration is valid. If it is not, troubleshoot with the help of your instructors, and try again to create the solutions.

Calibration Standards C2-F2 — serial dilution, automatic volumetric pipettes (pipettors)

One person in your team should create standard solutions for calibration set 2 (C2-F2) by serial dilution, using automatic pipettes:

1. Prepare solution "C2" using a 1.000 mL volumetric pipette, quantitatively transfer 1.200 mL of sodium acetate matrix to a clean tube. Then, add 1.800 mL of solution B and mix well. These steps can be accomplished by dialing the pipette to half of the desired volume, and making two careful transfers to double the volume.

Important points:

- Use the pipette to gently mix this solution by slowly pipetting up and down, but be careful not to "shoot" the solution up into the barrel of the pipette by releasing the plunger too quickly. If the solution goes up too high, it will become contaminated, and the pipette should be cleaned - consult your instructor if this happens and start over.
 - Note that the error for all transfers must be propagated!
 - Due to the small volume of this solution, **record the absorbance of the C2 solution before creating the next one.** See "Data Collection" instructions below.
2. Use the "Data Collection" instructions below to record the absorbance of solution C2. Then use solution C2 to create solution D2, and repeat this process for E2, and F2.
 - Use the pipette to mix each solution, then re-use your tip for the next solution.
 - Reserve a second tip exclusively for transferring the "clean" sodium acetate matrix.
 - Store all leftover solutions in labeled tubes and keep tubes closed to avoid evaporation.
 - Make sure to also collect the absorbance of the Blank, the Validation Standard, and the Sample.
 3. Use the data to determine if your calibration is valid. If it is not, troubleshoot with the help of your instructors, and try again to create the solutions.

Data Collection using a visible spectrometer

Measure and record the absorbance of each solution at 612 nm using a Spec 20. Instructions for operating the Spec 20 are in the section, [Operating Instructions for Spec 20 Visible Spectrometers](#).

Procedure for data collection

1. Blank the Spec 20 using a cuvette filled with 0.01M sodium acetate. Keep this cuvette filled with sodium acetate for future use as a blank.
2. Fill a second cuvette with 0.01 M sodium acetate and measure its absorbance to be sure the second cuvette matches the first. The reading should be very close to zero. Record the triplicate absorbance measurement in your data table - this is absorbance at concentration zero.

3. Measure and record the absorbances of each solution in the serial dilution.
4. Measure and record the absorbance of the "control" solution.
5. Measure and record the absorbance of the "unknown" solution.
6. Keep your solutions until you have validated your calibration curve and determined that you do not need to repeat any of the absorbance readings.

Tips for data collection

- Use **one** cuvette (a "working cuvette") for all measurements and keep a second separate cuvette for all blanks ("blank cuvette"). The blank cuvette should be filled with 0.01 M sodium acetate solution.
 - After the instrument is zeroed with the blank cuvette, measure the absorbance of the working cuvette after filling it with the 0.01 M sodium acetate solution - this is your absorbance at zero concentration of analyte.
 - Be very sure that both cuvettes are clean before making these measurements.
- Measuring sequence and working with the cuvettes:
 - Use the same working cuvette for all measurements, and the same blank cuvette for all zeroing.
 - It is essential that residual solution from the previous measurement does not dilute or contaminate your next sample!
 - You can clean and dry the cuvette between measurements - but this is not recommended. It is tedious and ineffective unless you have access to a cuvette washer. Do not stuff towels into a cuvette to dry it, and **never blow compressed air into a cuvette**.
 - To prepare the cuvette for the next measurement:
 - i. Empty the cuvette and tap the cuvette gently upside down on a soft towel to remove as much liquid as possible.
 - ii. Rinse the cuvette with a very small amount (a few drops) of the next solution to be measured, and do this three times.
 - iii. Add the bulk of the next solution and make the measurement.
 - If you are measuring multiple solutions sequentially, work starting with the least concentrated solution and proceed in order of increasing concentration. However, this is not possible using the automatic pipette protocol due to limited volume.
 - The outside of the cuvette should always be clean and dry.
- Take three measurements for each solution (use same liquid for each measurement; take cuvette out and wipe down then replace in instrument) and calculate the average absorbance and standard deviation from three replicate readings for each solution. Usually, you should get exactly the same absorbance reading for all three replicate measurements. The three readings should not differ by more than ± 0.003 absorbance units or 1% of the absorbance (see uncertainty associated with the spectrometer readings described in the operating instructions.)
- Between each measurement, the sample cuvette should be emptied and then rinsed twice with small (ca. 1 mL or less) aliquots of the next solution to be assayed.
***Work carefully because you have limited quantities of each solution.
- Between measurements, check the stability of the instrument by inserting the blank and making sure it still reads zero absorbance.

For each calibration, measure the concentration of the following solutions:

1. **Blank** (sodium acetate matrix)
2. **Each calibration standard** (work from lowest to highest concentration F→B)
3. **Validation standard** (determine whether the calibration is validated)
4. **Sample** (determine and report its concentration)

Each solution's absorbance should be measured three replicate times and recorded in a table in your notebook. Store all solutions in sealed containers until the second week of this module and discard them only after you are certain you no longer need them.

Important Note

Measure each absorbance three times. If the standard deviation of the replicate absorbances is much greater than the uncertainty (tolerance) expected for the instrument (see [Operating Instructions for Spec 20 Visible Spectrometers](#)), there may be a technical issue with the spectrometer!

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

Reflection Prompt **Runaway argument for \eatSpaces?** Comparing Tools for Calibration

Your team has now completed calibration curves using two types of solution preparation tools: Class A volumetric glassware and automatic pipettes. Use your data and your experience to **compare these tools** in terms of both **analytical performance** and **usability**.

In your response, address the following:

- **Precision and Accuracy:** Which tool gave more precise and accurate results? Use the the following as evidence to support your answer:
 - **calibration statistics** from each calibration fit (correlation coefficient, error in slope, error in intercept)
 - **calculated error in solution concentrations** (from error propagation calculations).
- **Convenience and Practicality:** Which set of tools was faster or easier to use? Which was more prone to mistakes or required more careful technique? Use your direct experience from the lab session to support your reasoning.

Your reflection should demonstrate that you can use both **data** and **scientific reasoning** to justify your calibration strategy decisions.

*Reflection prompts should be addressed in your lab notebook and should be included with the lab notebook submission.

Shut down\cleanup

Only when you are absolutely sure that you are finished for the day, please do the following shut-down and cleaning tasks:

- Store solutions in screw-top containers. Add parafilm around the seal of the cap. Be sure all containers are labeled with your name, date, section, and the contents of the container.
- Bromocresol green and sodium acetate waste should go in the **labeled waste container in the hood** dedicated for waste. Do not overfill the waste - notify a TA if the waste is above the shoulder of the bottle (where the bottle begins to narrow).
- Volumetric pipettes should be rinsed with deionized water to remove excess reagent solution, and then placed in the **pipette washer with tips UP (toward the ceiling)**. The pipette washer is a huge white plastic tube with a screen at one end. It will be in the sink near the drawers where the pipettes are stored.
- Rinse all other glassware 3 times with deionized water and remove all labels. Dry the glassware as well as possible and return it its storage location.
- Check the sink and other areas where you worked to be sure you did not leave anything lying around.
- Replace all automatic pipettors to their racks and place the tips nearby.
- Wipe down the bench and sink area with a damp paper towel.

Sources:

1. Skoog, D. A.; West, D. M.; Holler, F. J. Fundamentals of Analytical Chemistry, 7th edition, Saunders College Publishing, 1996.
2. Pickering, M. and Heiler, D. J. Chem. Ed., 1987, 64(1), 81.

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2.3: Week 2

Pre-Lab Assignment



Due: This pre-lab assignment is due prior to your Wet Lab meeting on the second week of classes.*

*The pre-lab assignments in this course are due prior to arriving at the Dry-Lab meetings — *except for this first module*. Due to the unique schedule in the first two weeks of the semester, the first two pre-labs will be due prior to arriving at Wet Lab meetings. Please note that starting the third week, all pre-lab assignments are due prior to the dry lab meeting (Monday)!



Purpose: The first part of this pre-lab is designed to help you reflect on your Week 1 and prepare you for experimental design in Week 2. This preparation will also support your ability to prepare a high quality post-lab report.



Week 2 Pre-Lab Tasks: Part 1

1. **Complete and submit any unfinished business from last week as part of this pre-lab submission.**
2. **Begin preparing your report:**
 - a. **Prepare a data table** that summarizes the concentration vs absorbance data for both calibrations.
 - Include propagated error for each standard's concentration. (Show at least one full worked example in your notebook).
 - Include average absorbance readings with standard deviation.
 - Do not include replicate absorbance readings - that belongs in your notebook, but not in a report.
 - b. **Prepare figures** to summarize your validated Session 1 data (the instructions here assume you have data to compare volumetric glassware to automatic pipettes).
 - Both curves plotted on the same axes (overlaid on the same plot).
 - Both regression models (lines) plotted.
 - Report the statistics for each linear regression.
 - c. **Discuss your results.**



Purpose: The second part of this pre-lab is to prepare you to engage in productive discussion and planning immediately upon arriving to the Week 2 Wet Lab session. You will need to discuss the plan with your teammates and come to consensus before beginning to work.



Week 2 Pre-Lab Tasks: Part 2

1. **Prepare data tables and a procedure for the following:**
 - **Priority 1:** Create a calibration using the opposite set of tools that you used last week. (Each person should try once with volumetric glassware and once with pipettes.)
 - **Priority 2:** Repeat your calibration from last week if you could not validate your curve.

Dry Lab Agenda

There is no Dry Lab meeting this week due to the Labor Day Holiday.

Wet Lab, Week 2: Comparing Strategies for Dilution (Serial vs. Individual)

Based on the results from part 1, and in consultation with your instructors, create new calibrations. Work with your team so that each person contributes one additional calibration curve to the dataset.



Notebook

Record your plan, your lab work, and your response to the reflection prompt in your notebook. Your notebook should be submitted on canvas prior to leaving Wet Lab.

Reflection Prompt 2.3.2: Comparing Dilution Strategies for Calibration

Each person has now completed calibration curves using two different sets of tools, and as a group, you have more replicates on which to base your judgement: Use your calibration data and your lab experience to evaluate these two sets of tools.

1. **Calibration Quality:** Were all the calibration curves (from Weeks 1 and 2) validated using the control solution? Are there any trends in accuracy and precision depending on the type of tool used?
 - Include discussion of whether the validation standard (control) result was within the expected range for each tool set.
 - Discuss any differences in R^2 , error in slope, and error in intercept between the two strategies.
 - Discuss any differences in propagated errors.
2. **Convenience of micropipettes vs volumetric glassware:** Were either of the methods easier or more convenient?
3. **Overall Method Recommendation:** If you were asked to design a calibration for a future experiment, which tools and which dilution strategy would you choose, and why? What parameters (eg time constraints, requirement for reproducibility) might change your decision?

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2.4: Post-Lab

This assignment is designed to help you practice presenting scientific results: you will practice writing a results section and writing a discussion section—two essential skills you will use throughout this course and in your future work as a scientist. In this post-lab, you will interpret data, communicate findings through clear figures and statistics, and reflect on your experimental design choices. These are the same skills you will apply in the three major projects later this semester.

This assignment is graded and considered final—revisions are not accepted. Use it as an opportunity to check your understanding of calibration, error analysis, and data presentation before we move into larger projects. Completing this assignment thoughtfully will help you identify strengths and areas for improvement before those higher-stakes assignments.

✓ Post-Module Assignment

1. **Results Section:** In your own words with data tables and plots created by you:
 - **Clear figures that show your calibration curves.**
 - Label each curve with the tool used (Class A glassware or automatic pipettes).
 - Overlay the curves for easy comparison.
 - **Clear Descriptions:** All figures should be introduced/described to the reader using text - just showing data is not sufficient.
 - State the key calibration statistics for each curve: For example, R^2 , slope $\pm$ error, intercept $\pm$ error.
 - State whether each calibration was validated using your control solution.
 - Report the calculated concentration of your unknown sample using each method.
2. **Discussion Section:** Write a short discussion (~300–500 words) that:
 - Compares the results: Which tools produced more accurate and precise data? Back this up with your statistics and your error propagation results.
 - Explains which method was easier, faster, or more practical for you in the lab.
 - States your recommendations to your future self: When would you choose Class A glassware vs automatic pipettes? What advice would you give yourself next time to design a calibration plan that works best for your goal?

📌 Note on Writing and Format of Post-Lab Assignments

Beginning with this module, all post-lab report sections should be written in the style of a scientific manuscript. For Module 2, this means you are responsible for writing complete **Results** and **Discussion** sections, with clear headings, figures, and tables formatted appropriately. Later modules will expand to include additional manuscript sections (Introduction, Methods, Conclusions, etc.), but you should treat each post-lab from this point forward as a piece of a professional scientific manuscript.

Reminders and Tips:

- Review the expectations for a figure (PChem Lab!) to be sure your data plots, tables, and their captions are meeting expectations.
- Use correct units and report values with appropriate errors and certainty (sig. figs. and errors).
- Describe each piece of data using text in the results section. Figure captions and data alone are not sufficient for a results section.
- Proofread your work for clarity.
- Think of this as writing a small piece of a scientific report for your future research self.
- Think of this as practice for writing the results and discussion sections for future assignments in this course.

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CHAPTER OVERVIEW

3: What's on Your Money? Designing Analytical Strategies for Trace Organic Residues

Learning Objectives

- Students will **describe the molecular-level models** of chromatographic separation and mass spectrometric identification that are relevant to their work.
- Students will **use models to explain** how chromatography and mass spectrometry separate and specifically identify trace organic material.
- Students will **use evidence from the literature** to design a protocol for sample preparation and a method to perform qualitative and quantitative analysis in the context of detecting and quantifying trace organic material.
- Students will **demonstrate competence with instrumentation and instrument software** to obtain and analyze chromatographic and mass spectral data.
- Students will apply chemical knowledge to **analyze and interpret chromatographic and mass spectral data** in order to identify compounds and quantify illicit substances on currency.
- Students will **communicate the broad relevance, specific goals, methods, results, and analysis** of their work in the form of a scientific manuscript.

[3.1: Introduction - Money](#)

[3.2: Week 1 - Money](#)

[3.3: Week 2 - Money](#)

[3.4: Week 3 - Money](#)

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3.1: Introduction - Money

What's in your wallet? If you're holding paper currency, you may also be holding trace amounts of illicit drugs. Studies suggest that up to 80% of U.S. banknotes carry measurable levels of cocaine—and some are contaminated with methamphetamine, oxycodone, or other organic residues. These substances are likely transferred through indirect contact, such as with contaminated counting machines or storage environments.

In this experiment, you'll investigate what's on your money. You will be responsible for designing your own sample preparation procedure and calibration strategy to detect and quantify organic contaminants on paper currency. You'll choose what compounds to target—such as cocaine, methamphetamine, or other trace substances—and apply analytical principles to decide which tools and techniques are best suited for isolating and identifying them. Your team will extract material from bills, analyze it using a sensitive detection method that supports compound identification through fragmentation data, and interpret your findings based on comparison with known standards and spectral databases.

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3.2: Week 1 - Money

Pre-lab Assignment:

 **Due:** This pre-lab assignment is due prior to the Week 1 Dry Lab meeting.

 **Purpose:** The first two questions in this pre-lab assignment are designed to engage you in thinking about how organic compounds, such as cocaine, interact with paper currency at the molecular level. By analyzing the chemical structures of both the contaminant and the currency material, you will apply your understanding of intermolecular forces to explain why some organic substances adhere strongly and persistently to surfaces like paper. These foundational chemical principles will help you understand the behavior of trace organic compounds in real-world forensic and analytical contexts.

Questions:

- What molecular-level interactions contribute to the adsorption of cocaine onto paper currency?** In your response, include the chemical structures of cocaine and cellulose (the primary component of paper). Draw a representation of how cocaine may interact with the cellulose surface. Identify and explain the intermolecular forces (e.g., hydrogen bonding, van der Waals interactions, ionic interactions) that are likely responsible for the adsorption and retention of cocaine on the surface of currency.
- Make a prediction about what organic compounds you will find on your currency during this experiment,** using evidence to support your prediction. Use Web of Science or another literature database to research substances that contaminate currency, and might be found on your money.

 **Purpose:** The second group of questions in this pre-lab assignment will help you practice using scientific literature to justify key decisions in analytical method design. By exploring how researchers select instruments, prepare samples, and use internal standards, you will learn how to propose a method that balances accuracy, safety, cost, and practicality. These skills will support your ability to justify experimental choices in your scientific manuscript.

Questions:

Use a research database to access the articles cited below and add them to your Zotero Library. Use these papers as reference for selection of your analytical methods and sample preparation protocols. *You do not need to read these papers from start to finish. Read the sections of the papers that help you with this assignment.*

Papers:

- Parker, Patrick D., Brandon Beers, and Matthew J. Vergne. "What Is in Your Wallet? Quantitation of Drugs of Abuse on Paper Currency with a Rapid LC-MS/MS Method." *Journal of Chemical Education* 94, no. 10 (2017): 1522–26. <https://doi.org/10.1021/acs.jchemed.7b00196>
- Negrusz, A, JL Perry, and CM Moore. "Detection of Cocaine on Various Denominations of United States Currency." *Journal of Forensic Sciences* 1972-2005 43, no. 3 (1998): 626–29. <https://doi.org/10.1520/JFS16193J>.

1. What instrument will you use?

- What kind of instrument detector enables specific identification and quantitation of trace organic analytes?
- What kind of instrument component can separate compounds in a mixture so that they can be analyzed by the detector described in (a).
- Propose an instrument that you will use to detect and quantify trace organic compounds on your currency. Justify why this instrument is the right choice for specific identification and quantitation of very low concentrations of organic analytes.

2. **Suggest a procedure to prepare the sample;** the protocol (1) will extract organic material from currency, (2) is simple and low cost, (3) avoids the use of particularly hazardous solvents (like dichloromethane or chloroform), (4) takes less than 30 minutes to prepare a sample, and (5) does not destroy the currency.

3. In the two manuscripts above, **what internal standards were used** during calibration and quantitation?

4. **Will you use an internal standard?** Decide whether you think an internal standard is necessary. Justify your answer with evidence from the papers or from other reliable sources.

Dry-Lab Group Discussion:

In this discussion, you will collaborate with your research team to refine your understanding of how organic compounds interact with paper currency and you will refine your extraction protocol and analysis method. By drawing molecular interactions, evaluating extraction protocols, and discussing calibration strategies, you will apply chemical principles and literature evidence to make informed decisions about your experimental approach. This discussion should will be facilitated in small groups (2-4 students) by a member of the instructional team.

Instructions:

- Begin the session by grouping up and comparing the answers to your pre-lab questions in your group for ~5 minutes.
- Assign roles (scribe, reporter, spokesperson, time keeper)
 - The Time Keeper will start a timer and encourage group consensus.
 - The Scribe will record the group work.
 - The Spokesperson will summarize your group's discussion and questions to the instructors.
 - The reporter will submit the group record at the end of class.
- Write your work on the board and try to come to consensus as a group.
- If you have trouble reaching consensus, consult another group for advice, or consult an instructor when they are available.

Goals for Dry-Lab discussion:

1. Team consensus about method choice: Which instrument is best for this application, considering availability of instruments in the undergraduate learning labs?
2. Team consensus about extraction protocol: How will you prepare the sample?
3. Team consensus on calibration strategy: Will you use an internal standard?

Group Discussion Prompts

1. **Predict the relative amounts of cocaine and methamphetamine you will find on your currency.** Using your pre-lab response as a starting point, draw on the board:
 1. A molecular-level picture of how cocaine is adsorbed onto cellulose-based currency. Include chemical structures and show the intermolecular interactions.
 2. Do a similar analysis for methamphetamine and cellulose. Do you expect methamphetamine to adsorb onto cellulose as well as cocaine does? Justify your prediction using the molecular structures as the basis.
 3. Predict the relative amounts of methamphetamine and cocaine on paper currency.
2. **What instrument will you use, and what are the appropriate settings?** Consider the instruments to which our course has access, and use the literature methods as a starting place. How will you need to modify instrument settings to see them maximum number of possible contaminants?
3. **Discuss your proposed extraction protocol with your group,** consider the limitations stated below, and finally consult your instructors. Discuss what are the costs and benefits of the suggested approaches and come to a group consensus about what protocol your team will use.
 1. Limitations of the analytical method: For example, if using gas chromatography, the sample should be dissolved in a solvent with a low boiling point so that the solvent elutes quickly and does not interfere with detection of the analyte ions. If using liquid chromatography, common mobile phases are water, methanol, and acetonitrile.
 2. Limitations of analytical solution preparation: If the solvent is too volatile, it will be challenging to transfer quantitatively, and difficult to prevent evaporation (which would affect concentration).
 3. Limitations imposed by purchased standards: Since we cannot purchase these controlled substances without a license, the standards are purchased as dilute solutions. The standard solutions (cocaine, methamphetamine) are only available as dissolved in Methanol - this solvent is versatile and suitable for both GC and LC applications.
 4. Solubility: The solutes must be soluble in the solvent.

5. **Safety:** Some solvents are not only flammable, but also particularly toxic. Do not use solvents that are particularly toxic, such as chlorinated solvents (chloroform and dichloromethane).
4. **Discuss your proposed calibration method with your group.** Specifically, discuss whether you think an internal standard should be used in calibration, and justify your response using evidence from analytical principles and the literature. Please consider the limitation that deuterated analytes are typically expensive.
5. **Write detailed protocols** for sample extraction and calibration strategy in your notebook to prepare for the wet-lab session.

Wet-Lab Agenda:

1. **Lessons and Demonstrations:** Your TA will give a short lesson about the instrument you will be using for analysis. Draw diagrams of the instruments and equipment in your notebook.
 - Instrument: Draw a diagram of the instrument components in your lab notebook. Include and label the autosampler, sample, injector, the components of the chromatography system, and the components of the mass spectrometer.
 - Syringe-Driven Filter: Draw the syringe driven filter device in your notebook.
 - Autosampler vials: Draw a diagram of the autosampler vial.
2. **Sample Preparation:** Confirm that all group members have the same plan for sample extraction. Record any observations and deviations from the planned protocol in your notebook and perform the sample extraction.
3. **Instrument method:** Choose or write an appropriate method for analysis of your samples. Record the details of the instrument settings in your notebook.
4. **Prepare sequence for sample injection.** Use the instructions in the instrument protocol to create a sequence file to inject the following:
 - A blank (the solvent matrix of your samples)
 - Sample(s)
 - A standard mixture
5. **Group discussion about electron ionization mass spectrometry.** (while waiting for samples to run)
 - Search for cocaine mass spectrum using the NIST WebBook.
 - Identify mass ion (303), and two most abundant fragment ions (82, 182) of cocaine.
 - Explain how electron ionization created the mass ion and the fragment ions (all are $z=1$).
6. **Prepare sequence for injection of calibration standards** to run overnight.
7. **Export qualitative data figures and report.**
 - Create a table that lists the compounds found, their mass ions, retention times, and confidence in the assignments. An example table is below.

Example Table:

Compound	Ions identified	Found r.t. in standard (min)	Found r.t. in sample (min)	Confidence in compound identification
exampleamine	42, 84	3.42	3.43	94%
examplone	54, 154, 187	5.43	5.42	85%
methylexample	56, 74	Not present	4.38	80%

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

📌 Reflection Prompt **Runaway argument for \eatSpaces?**

1. Briefly describe the extraction procedure your group developed. Explain the reasoning behind your choices, including how these choices can be justified using chemical principles and/or literature sources.
2. Justify your group's decision about whether to use an internal standard in your analysis. If you chose to use one, explain how its chemical properties make it appropriate for this experiment. Support your reasoning using relevant chemical principles (e.g., structural similarity, retention time, ionization efficiency, or extraction behavior).

*Reflection prompts should be addressed in your lab notebook and should be included with the lab notebook submission.

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3.3: Week 2 - Money

Pre-lab Assignment

Due: This pre-lab assignment is due prior to the Week 2 Dry Lab meeting.

Purpose: This pre-lab is designed to prepare you for a productive dry-lab discussion and upcoming data analysis. By organizing your qualitative results and drafting your Methods section now, you'll be ready to share work-in-progress and get early feedback to improve your report.

Follow up from Week 1

1. **Complete and submit any unfinished business from the last meeting.**
2. **Begin preparing your manuscript document:**
 - a. **Complete a draft of your Materials and Methods section.**
 - b. **Complete the Qualitative Results section, including description and figures that show the results of qualitative analysis.**
 - Prepare a table of compounds identified in your sample.
 - Prepare a figure that provides data to support the identification of components of your sample.
 - The table and figures should complement each other.

The exercise below uses example data to help you practice choosing between linear and nonlinear regression models using quantitative evidence. This practice will prepare you to analyze your own calibration data in lab this week and make informed, data-driven decisions. Completing it in advance will help you fully engage in the upcoming lesson on quantitative analysis of GC-MS results.

Questions

Practice evaluating regression models for calibration data.

3. **Examine the example data sets below and determine whether a linear or nonlinear regression model is more appropriate in each case.** For each data set:

- Fit both a linear and a nonlinear model (e.g., quadratic or exponential, as appropriate)
- Compare the quality of the fits using appropriate error metrics (e.g., R^2 , residual plots, standard error, or percent error in calculated concentrations)
- Justify your choice of model using quantitative evidence, not just visual analysis.

What patterns in the data would indicate that a nonlinear model may be more appropriate than a linear one? How would an inappropriate regression model affect your interpretation of an unknown?

Data Set A		Data Set B	
Conc. (ng/mL)	Response	Conc. (ng/mL)	Response
0.1	15	0.1	15
0.5	81	0.5	81
1.0	168	1.0	132
2.5	430	2.5	398
5.0	957	5.0	710
10.0	2397	10.0	1551
15.0	3986	15.0	2321
20.0	6384	20.0	2970

Hint

An appropriate analysis of fit should include a plot of residuals. You've practiced this before in your P-Chem coursework—reviewing your residual plots from the FTIR of HCl and the Apples Lab (Matlab scripts) may help refresh your memory and guide your approach here.

Dry Lab Group Discussion

In this session, you will work with your research team to improve your qualitative results section, and prepare for quantitative analysis of your calibration data. You will also share drafts of your Materials and Methods section to get early feedback from your peers. By analyzing example calibration data and refining your own data presentation, you will deepen your ability to justify decisions using chemical reasoning and statistical evidence. This discussion will take place with the whole team, and will be facilitated by a member of the instructional team. Please work with your team to follow the instructions below.

An instructor will come address questions as soon as they are available; if for some reason an instructor cannot help you during the session, we will address any questions reported out by your reporter before the end of the day.

Instructions:

- Assign the following roles, then follow the agenda below.
 - The recorder should record the conversation and questions on the board or in a shared electronic document.
 - The time keeper should set a timer with a loud noise to keep the group work on track.
 - The reporter will submit a photo of the white board or the shared document at the end of the period.

Agenda:

1. Discussion about how to interpret and show Qualitative Results - Use the Group Discussion Prompts to engage in discussion.
 - Time: **15 minutes**
 - Record: main points and questions that you have.
 - Prepare to discuss these points with your instructor when they become available.
2. Discussion about regression analysis - Use the Group Discussion Prompts to engage in discussion.
 1. Time: **15 minutes**
 2. Record: main points
 3. Prepare to discuss main points with your instructor.
3. Peer Review: Begin the session by peer-reviewing another person's document draft. Review the Materials and Methods Section and Qualitative Results section, adding constructive and critical feedback.
 - Time: **15 minutes**
 - Please be silent during this time so others can complete the assignment.
4. Discuss Materials and Methods - what questions came up as you reviewed other's work?
 - Time: **15 minutes**
 - Record: main points and questions on the board.

Group Discussion Prompts

Analysis of Qualitative Data:

1. Confirmation Through Multiple Lines of Evidence

- If two compounds have similar retention times, what additional evidence can confirm identity?
- How can you use both the retention time and fragmentation pattern to confidently assign identity to a peak?

2. Library Search Validation:

- What does the library match score tell you, and how reliable is it?
- How can comparing the MS of an unknown directly to the MS and retention time of a standard help confirm or challenge the library ID?
- What should you do if a peak in your sample does not match any in your standard?
- How would you communicate uncertainty in compound identification in your report?

Regression Analysis:

The form of a linear and a quadratic regression model is given below. Inspect these models:

$$\text{Straight Line: } y = mx + b \quad (3.3.1)$$

$$\text{Quadratic: } y = ax^2 + mx + b \quad (3.3.2)$$

The quadratic model includes more coefficients than a linear model, which gives it greater flexibility to fit the data—even if the underlying relationship is actually linear. This can lead to overfitting and the false impression that the quadratic model is "better." Even when a quadratic model is appropriate over a wide range of concentrations, a linear model might be more appropriate if your sample is at the lower end of the range, within a linear portion of the curve.

3. **Which statistical measures might mislead you into thinking a more complex model is better?**
4. **How can you know which to use?** What additional evidence can help you decide which model is truly more appropriate for your calibration data?

Goals for Dry Lab:

- Deepen your understanding of qualitative data:
 - Results section: Ensure your chromatograms and mass spectra are labeled and ready to use in your report, and that your description of the results is appropriate.
 - Discussion section: Prepare a strategy to write the Discussion of qualitative results.
- Improve your Materials and Methods section: Identify and address any gaps or unclear steps before you move forward with writing.
- Deepen your understanding of Calibration: Review how to evaluate regression model fit so you can apply those skills to your own calibration data later this week.

Wet Lab Agenda

1. **Access your data:** You initiated the calibration in Week 1. You will need to know your file names and in which directory you stored your data so you can locate the data and complete the analysis.
2. **Lessons and Demonstrations:** Your TA will give a short demonstration on how to perform quantitative analysis using the Agilent Mass Hunter Quantitative Analysis software
3. **Perform Quantitative Analysis:** You will work in groups to create a *batch file* and *method* for quantitative analysis
 - As a group, you will deliberate on the use of a straight line vs. quadratic regression model in order to most accurately represent your data. *In your lab notebook, make note of the results of any statistical or regression analysis used to come to a decision*
 - Produce a calibration curve using the regression model selected.
 - Perform quantitative analysis on your samples and standards. *Make sure to include error analysis!*
4. **Continue writing** the *Results* and *Discussion* sections of lab report (only after completing the reflection prompts below.)

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

Reflection Prompt **Runaway argument for \eatSpaces?**

- **What regression model did you choose as the best fit for your calibration data?** Justify your choice using appropriate evidence.
- **In the sample data, how do the amounts of cocaine and methamphetamine compare to one another?** Rationalize this observation considering the molecular structures of cocaine, methamphetamine, and cellulose. You may also speculate about other factors that affect the presence of these substances on currency, as you see fit.

*Reflection prompts should be addressed in your lab notebook and should be included with the lab notebook submission.

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3.4: Week 3 - Money

Pre-lab Assignment

 **Due:** This pre-lab assignment is due prior to the Week 3 Dry Lab meeting.

 **Purpose:** This pre-lab is designed to prepare you to share work-in-progress and get early feedback to improve your report.

Follow up from Week 2

1. **Complete and submit any unfinished business from the last meeting.**
2. **Prepare your manuscript document:**
 - a. **Update the Materials and Methods section** based on last week's peer feedback.
 - b. **Update the Qualitative Results section** based on last week's peer feedback.
 - c. **Write your Discussion of Qualitative Results.**
 - d. **Write the Quantitative Results section, including description and figures that show the results of quantitative analysis.**
 - Prepare a table that communicates the results of calibration and quantitation.
 - Prepare one or more figures that shows calibration data.
 - The table and figures should complement each other.
3. Bring a **printed copy** of your manuscript to the Dry Lab meeting.

Dry Lab Group Discussion

In this session, you will peer review each other's reports and practice helping each other by giving impactful, actionable feedback.

Wet Lab Agenda

Record all work in your lab notebook as usual. If you are not doing wet-lab work in this session, please keep still notes of your writing-related conversations in your lab notebook and complete the reflection prompts as usual.

1. **Revise Manuscript and Write Discussion, Conclusions, and Introduction Section.**
2. **Meet with Writing Consultants.**

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

Reflection Prompt **Runaway argument for \eatSpaces?**

1. **How has writing about your lab experience supported your learning?**
2. **How has working with the EWC's supported your writing and learning?**
3. **What have you done that has supported your learning in this module? What could you do better in the future?**
4. **What were the ways in which your instructional team supported your learning? What could they do better in the future?**

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CHAPTER OVERVIEW

4: Quant-Tea-fy It! How much caffeine is in your drink?

Learning Objectives

After completion of this module you will be able to:

- **Describe and use molecular-level models** of electronic absorption spectroscopy and chromatographic separation to explain how analytes are detected and separated, including limitations of each method.
- **Apply chemical knowledge and instrumental theory to design, justify, and evaluate** quantitative analyses of mixtures using UV-vis spectroscopy and HPLC.
- **Demonstrate reproducible preparation of calibration standards and justify choices** of techniques that ensure accuracy in quantitative analysis.
- **Choose and evaluate calibration models**, using calibration statistics and estimated uncertainty to interpret results and assess method suitability.
- **Demonstrate competence with instrumentation and software** by selecting appropriate operating parameters, troubleshooting data collection, and critically analyzing spectroscopic and chromatographic output.
- **Interpret and integrate spectroscopic and chromatographic data** to identify analytes and determine their concentrations in complex mixtures.
- **Communicate** the significance, goals, methods, results, and interpretations of the work in a professional scientific manuscript format.

This lab manual includes a brief overview of HPLC. For fuller coverage of the topic, refer to the textbook for CHEM 401. Libretexts also has a similar text (see [Ch 12 of Analytical Chemistry 2.1](#) (Harvey)).

[4.1: Introduction - Tea](#)

[4.2: Week 1 - Tea](#)

[4.3: Week 2 - Tea](#)

[4.4: Week 3 - Tea](#)

[4.6: TA Notes](#)

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4.1: Introduction - Tea

Caffeine is a naturally occurring alkaloid found in coffee, tea, chocolate, and many energy drinks. It is one of the most widely consumed psychoactive substances in the world, valued for its stimulating effects on the central nervous system.^{1,2} While moderate caffeine intake is generally considered safe for most adults, excessive consumption can lead to health concerns such as insomnia, elevated heart rate, and increased blood pressure.³ Regulatory agencies, including the U.S. Food and Drug Administration, recommend labeling caffeine content for certain products, making accurate quantitation an important public health and consumer information goal.⁴

In this module, you will investigate the caffeine content of beverages using two complementary analytical techniques: UV–visible spectroscopy and high-performance liquid chromatography (HPLC). UV–vis allows for rapid, inexpensive quantitation of caffeine in relatively simple matrices, while HPLC provides high selectivity and sensitivity for complex mixtures. Together, these methods illustrate how analytical chemists select and validate techniques based on the properties of the analyte, the complexity of the sample, and the accuracy required.⁵

You will prepare and analyze calibration standards, validate your methods using control samples, and apply your results to determine caffeine concentrations in real beverage samples. Along the way, you will connect molecular-level models of light absorption and chromatographic separation to the data you collect, gaining both conceptual understanding and practical skills in quantitative chemical analysis.

References

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4.2: Week 1 - Tea

Pre-Lab Assignment

 **Due:** This pre-lab assignment is due prior to the Week 1 Dry Lab meeting.

 **Purpose:** Prepare for the Week 1 Dry Lab meeting and Wet Lab session by selecting your beverage, planning your calibration strategy, and collecting background information for key parts of your scientific manuscript.

Pre-Lab Tasks for Week 1

1. Select a Beverage & research its caffeine content:

- Choose a caffeinated beverage to analyze (each group member should pick at least one unique beverage).
- Research its approximate caffeine concentration using reputable sources (e.g., USDA FoodData Central, peer-reviewed publications, manufacturer data).
- Save the source(s) to your Zotero library and cite in ACS style.

2. Draft Introduction (partial)

- Write 1–2 paragraphs of background for your Introduction section. Include:
 - Brief explanation of caffeine's significance (chemical properties, psychoactive nature, health considerations), with citations.
 - Why accurate caffeine quantitation matters (public health, labeling regulations, quality control), with citations.
 - Include at least three references in ACS style.

3. Plan Calibration & Sample Dilution

- Select the tools (glassware, pipettes) and create a dilution plan for creating caffeine calibration standards in the *approximate* range **0.005 – 0.04 g/L** (the range does not need to be exact).
- Determine the dilution factor needed to bring your beverage sample into the middle of your calibration range. Your highest calibration point should be between 1.5 to 3 times the concentration of your sample.
- Present your plan in a clear, stepwise outline or table.

4. Prepare a Data Table Template

- Create a table to record your UV-Vis absorbance readings for calibration standards and beverage samples.
- Include columns for:
 - Sample ID
 - Caffeine concentration (g/L)
 - Absorbance at λ_{max}
 - Replicate #
 - Notes/Observations

Submit your pre-lab assignment on Canvas, and be prepared to justify your plan for solution preparation to your team.

Dry Lab Group Discussion

The purpose of today's Dry Lab is to share and critique your individual pre-lab plans for calibration and sample preparation. By the end of this session, you should have considered multiple approaches, anticipated challenges, and refined your own plan for Wet Lab. You will also begin drafting sentences for the *Methods* section of your manuscript.

Instructions:

- Begin the session by grouping up and assigning roles (scribe, reporter, spokesperson, time keeper)
 - The Time Keeper will start a timer and encourage group consensus.
 - The Scribe will record the group work.
 - The Spokesperson will summarize your group's discussion and questions to the instructors.
 - The reporter will submit the group record at the end of class.

Goals:

1. Refine your calibration plan and sample dilution plan so you can start preparing solutions right away in Wet Lab.
2. Make a plan for dealing with technical issues during data collection.
3. Begin writing the Methods section of your manuscript.

Group Discussion Prompts

1. **Calibration Choices:** What choices did you each make when designing the plan for your calibration standards?
 - What choices did you each make when designing calibration standards (concentration range, number of points, dilution method)?
 - What tools did you select, and why?
 - How much of each standard will you each produce? Will your volumes provide enough material for UV-Vis (~3 mL), HPLC (~2 mL), and filtering?
 - Share your plans for creating the calibration standards. Is the strategy reasonable considering the necessary volumes and the tools available?
 - After discussion, what adjustments (if any) will you make?
2. **Sample:** Each group member should choose a unique sample. No two group members should use the same *exact* beverage.
 - Why do we dilute beverage samples instead of analyzing them directly?
 - Compare each member's dilution strategy for the chosen beverage:
 - Is it reasonable?
 - Is the anticipated diluted sample concentration within the calibration range?
 - What challenges might arise with your particular beverage (e.g., temperature, carbonation, particulates, solubility)?
3. **Validation & Troubleshooting**
 - What will you do if your standards give absorbances outside the quantitative range for UV-vis (0.2–0.8 au)?
 - Are shorter path-length cuvettes realistic to use, or is further dilution more practical? Do you need to collect UV-vis data on every standard to confirm accurate sample preparation?
 - Do you have enough solution to accommodate repeats or adjustments?
4. **Data Collection and Sharing**
 - Please see the canvas page describing the **CHEM401L Data Drive** (CHEM401L Data on Google Drive).
 - How will your group share data files (UV-Vis scans, calibration results) to make sure everyone has multiple examples for their reports?
 - What naming conventions or folders will you use to avoid losing or mixing up files?
5. **Writing:** While this discussion is fresh in your mind, write a few sentences that can support the Methods section of your manuscript.
 - Draft one sentence describing your calibration preparation.
 - Draft one sentence describing your beverage preparation.
 - Draft a one-sentence description of how you will validate your calibration.

Wet Lab Agenda

Goals: Prepare a set of caffeine calibration standards using your preferred method and tools, and validate your calibration solutions using UV-vis. Also prepare and measure the absorbance of a beverage sample.

Do not throw away any solutions until the very end of this 3-week module! Plan to **store your solutions in sealed vials in the refrigerator for the entire 3 weeks.*

Prepare the UV-vis

- Turn on the UV-Vis if it is not already on.
- After an appropriate warm-up period, [Validate](#) the instrument.

Prepare Caffeine Calibration Standards

Materials required:

- Caffeine (99% powder)
- Ultrapure water
- Volumetric glassware
- Caffeine Validation Control solution (provided by TAs)
- Equipment for preparing solutions

You have access to:

- volumetric flasks to contain of 5, 10, 25, 50, 100, 250, 500, and 1000 mL
- volumetric pipette to deliver volumes of 1, 2, 3, 4, 5, 10, 15, 20, 25, 50, and 100 mL
- automatic pipettors to deliver between 100-5000 mL

A. Prepare a stock solution of 0.1 g/L caffeine

1. *Quantitatively* weigh out between 8.0 and 10.0 mg of caffeine. The caffeine can be found on the shelf near the weigh station area.
2. *Quantitatively* transfer the caffeine into a clean 100 mL volumetric flask. Rinse the weigh boat with ultrapure water to be sure all powder is rinsed into the flask.
3. Dilute to the volumetric mark with ultrapure water. The stock solution will have a final concentration of ~ 0.1 g/L. Calculate the actual concentration based on the mass of caffeine used.

Warning

There is a very small amount of caffeine powder, but it takes some time to dissolve completely. Be sure that caffeine is completely dissolved in about 50 mL of water before diluting to the volumetric mark and before proceeding to the next step. There is a sonicator in the lab that you can use to help break up small caffeine particles and aid in dissolution.

B. Prepare calibration standards

Write out a specific plan for doing this before you get to lab (see pre-lab assignment). Carry out a serial dilution to obtain a series of calibration standard solutions that are approximately evenly spaced (in terms of their concentrations). The highest concentration should be about 0.04 ± 0.01 g/L, and the lowest concentration solution should be approximately 0.005 ± 0.001 . Make enough of each solution so that you have *at least* 10 mL of each solution to play with. You should use ultrapure deionized water to make all solutions and dilutions. Ensure adequate mixing in each diluted solution before using it to create the next.

Prepare the Beverage Sample(s)

Each student should bring about 50 mL of a unique caffeinated beverage sample for testing. Beverages used in lab should not be consumed.

Store beverage solutions at 4° C

Beverages and beverage samples must be stored in the refrigerator to avoid microbial growth. Warm all liquids to room temperature before measuring volumes!

Dilute the beverage samples to reach a concentration that can be quantitated by your calibration range (based on the expected caffeine content of the beverage, and within the range of the calibration standards).

Example beverage preparations:

1. A Coffee, which is estimated at ~ 0.4 g/L caffeine.

- a. Pipette 5 mL of coffee into a clean and dry 100 mL volumetric flask and dilute to the mark with clean deionized water.
- b. Filter the sample using the provided filter.
 - First, rinse the filter three times with 3 mL clean deionized water.

- Rinse the filter with the sample solution by filtering the first 1-2mL of the sample into the waste beaker.
- After rinsing the filter apparatus with sample solution, filter the sample directly into a clean vial and label the vial.

2. A Tea, which is estimated at ~ 0.2 g/L caffeine.

- Pipette 10 mL of tea into a clean and Dry 100 mL volumetric flask and dilute to the mark with clean deionized water.
- Filter the sample using the provided filter.
 - First, rinse the filter three times with 3 mL clean deionized water.
 - Rinse the filter with the sample solution by filtering the first 1-2mL of the sample into the waste beaker.
 - After rinsing the filter apparatus with sample solution, filter the sample directly into a clean vial and label the vial.

3. A Soft Drink Sample, which is estimated at ~ 0.1 g/L caffeine.

- If the soft drink you selected is carbonated, decarbonate the soft drink by pouring it back and forth between two beakers until the bubbles cease.
- Pipette 10 mL of the soft drink into a clean and Dry 50 mL volumetric flask and dilute to the mark with clean deionized water.
- Filter the sample using the provided filter.
 - First, rinse the filter three times with 3 mL clean deionized water.
 - Rinse the filter with the sample solution by filtering the first 1-2mL of the sample into the waste beaker.
 - After rinsing the filter apparatus with sample solution, filter the sample directly into a clean vial and label the vial.

Analysis of caffeine with UV-vis

At the beginning of the lab period, check to confirm that the UV-vis is switched "on". The Cary 100 instruments have tungsten and deuterium lamps that must be warmed up for approximately 1 hour prior to use. The Cary 60 instrument has a xenon flash lamp, which does not require warm up. Be sure you discuss which instrument you will use with your TA, and if you are using a Cary 100, be sure it is on for 1 hour prior to use.

Validate your calibration solutions:

Your calibration solutions consist of caffeine dissolved in a simple matrix — ultrapure water. This fact makes specific detection of caffeine straightforward using a UV-visible spectrometer. For reasonable quantitative measurements, the absorbance readings should be between 0.2 - 0.8 au. If absorbance readings are outside of this range, measure using a different path length cell or dilute/concentrate the sample.

Using the Agilent Cary UV-vis spectrometer

Refer to the [instrument manual](#) for operation instructions — first [Validate](#), then set up a method to collect a [Scan](#) of each solution.

Depending on the instrument accessories and the number of samples you plan to run, the procedure below could vary slightly. Make sure to consult your instructors about any changes to this procedure.

- Using the [Scan](#) program to collect a baseline spectrum.
 - You will need either one or two cuvettes containing a sample blank (ie no analyte). The cuvette should not be empty; rather it should contain the background solution of your sample, but with no analyte. (In the case here, it should be ultrapure water).
 - Put the cuvette with a blank solution into the appropriate position(s). Then press the "Baseline" button on the left of the Scan window.
- Collect a scan of the blank to ensure it looks as expected. (Press "Start" to collect a scan.)
- Specify the file name and folder location where the data should be saved.
 - All data should be saved in the directory **C:\Data\CHEM401L\ within a folder for your group**. Make a new folder within the CHEM401L directory if one does not already exist.
 - Your file names should follow the convention "YYYYMMDD_Initials_###". In your notebook, you should keep a log of file names and a description of what is contained in each file. Prepare a table in your notebook for recording this information. Example is below (this info is made up!):

Data Table - File Location is C:\Data\CHEM401L\OurGroup

--

File Names: 20230811_KH_##	Absorbance at λ_{max}	Solution components	Comments
20230811_KH_01	0.0001	ultrapure water	This was just to check that the blank is as expected (ie a flat line)
20230811_KH_02	0.246	coffee sample	diluted coffee sample
20230811_KH_03	0.0809	0.01 g/L caffeine in ultrapure water	calibration standard 1...

- Remove the blank, then empty the cuvette.
- Place the most dilute calibration standard in the cuvette:
 - First, rinse the entire inside surface of the cuvette 3 times with small portions of the sample.
 - Place the sample in the rinsed cuvette.
- Place the cuvette with the sample in the appropriate position(s) in the instrument, and press "Start" to collect a spectrum of the sample.
- Repeat steps 5 and 6 above for each sample listed below. If you use the same cuvette and instrument for each, you do not need to repeat the baseline step. However, you should ensure that each calibration solution spectrum is at zero absorbance near 400 nm. If not, check with your instructors.
 - each calibration solution
 - the validation control solution
 - the diluted beverage sample
- After you have collected all data, it is helpful if you "save as" all of your data in one "batch" .csv file. To do this, choose "file" and then "save as" and choose ".csv". Give the file a name like "YYYYMMDD_Initials_all" to indicate it has "all" your data. Open this file using excel and check that it in fact contains all the spectra that you collected.

Check your data files and transfer to your own device:

Before you start cleaning up, check to be sure that you can find your data, and that they are in fact saved as .csv files. Also check that you can open these files and they include all your data as expected. Transfer your data files to the CHEM401L Drive folder (use the CHEM401L Drive shortcut on the desktop) and then to your own computer so you can work with them later.

Check following:

- You have YOUR data (not someone else's)
- You can open the files on your computer
- You know what is in the files and how they are structured
- You have a plan for how to plot the spectra and create the calibration curve from the data

Prepare for HPLC Analysis

As soon as you validate your solutions, prepare your calibration standards and beverage samples for the HPLC autosampler and set up a sequence to run your calibration standards, the control, and your samples on the HPLC.

Transfer solutions to autosampler vials

After validating your calibration series (the caffeine calibration standards), filter your solutions, and transfer them to autosampler vials, sealed well to prevent evaporation.

- Filter the solutions using a syringe and syringe filter. These can be found in the labeled drawers in the lab.
 - Use only one filter and one syringe for the entire series of solutions.
 - The filter and syringe can be reused if the solutions are filtered from low concentration to high concentration
 - Rinse the filter with the solution to be filtered:** Filter approximately 1 mL and dispose of it into a waste beaker.

- After the filter is rinsed, use it to filter the solution directly into the appropriate HPLC autosampler vial. (Figure 4.2.1)
DO NOT put tape on the vials; use a permanent marker to write directly on the vials.



Figure 4.2.1: HPLC/CE vials. The HPLC can also use plastic vials that look quite similar. Image is taken from www.thermofisher.com

Preparing a Sequence for HPLC analysis of the Caffeine Standards and Samples

In Week 1, you will *prepare* and *save* a sequence template, but you will not run it yet. A sequence is a list of instructions that tells the autosampler robot what samples to inject, how much, how to run the sample, and what to do with the data. Follow the instructions below to create the sequence template file that you will use next week.

- Place vials in an auto sampler tray
 - Do not move another group's samples.
 - Put standard caffeine solutions in slots 1-5 of one LETTERED row of Tray 1 (P1), with the least concentrated in slot 1 and the most concentrated in slot 5.
 - Place the beverage samples in slots 6-9 of the same row
 - Place a blank (a vial with ultrapure water) in row A
 - Place a control in row A
 - Take note of the position of each of your samples, standards, blank, and control.
- Record the positions of each vial in your notebook - Store the tray and samples in the refrigerator.
- Work with your TA and your group members to decide whether you will collaborate on one sequence for the whole group, or whether you will each create your own sequences, then run the group pf sequence templates as part of a Que.
- Set up a sequence ([See instrument manual](#)). An example sequence for one set of calibrations, a control, and four samples is shown below:

Sequence Table: HPLC 1

Line	Sample ...	Sample Name	Method Name	Injection Source	Injection Vol...	Inj/Loc	Sample Type	Cal Level	Update RF	Update RT	Cal Inte...
1	P1-A1	Blank	default	As Method		1	Blank				
2	P1-D1	Cal 1		As Method		1	Calibration	1	Average	Average	
3	P1-D2	Cal 2		As Method		1	Calibration	2	Average	Average	
4	P1-D3	Cal 3		As Method		1	Calibration	3	Average	Average	
5	P1-D4	Cal 4		As Method		1	Calibration	4	Average	Average	
6	P1-D5	Cal 5		As Method		1	Calibration	5	Average	Average	
7	P1-A1	Blank		As Method		1	Blank				
8	P1-A2	Control		As Method		3	Control Sample				
9	P1-D6	Coffee		As Method		3	Sample				
10	P1-D7	Tea		As Method		3	Sample				
11	P1-D8	Coke		As Method		3	Sample				
12	P1-D9	Diet Coke		As Method		3	Sample				
13	P1-A1	Blank		As Method		1	Blank				
14											

Run OK Cancel Apply

Figure 4.2.2: An example sequence table (Kathryn Haas; CC-NC-BY-SA 2025)

- Each line of the sequence table corresponds to a sample injection. Add as many lines as injections you wish to make.
 - Run one or two blank injections first (always!!)
 - Run your calibration series in order of increasing concentration

- Run a blank between your highest concentration calibration and the next sample
- Run triplicate injections of the control (set in Inj/Loc column)
- Run triplicate injections of the samples (set in Inj/Loc column)
- Set the Sample Type for each injection
- Set the injection volume to be consistent (or enter nothing and it will be defined by the method)
- You have not yet chosen a method, so leave the method name blank
- Press **OK**
- **Save** your Sequence Template using the Sequence menu from the top toolbar.

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

Reflection Prompt **Runaway argument for \eatSpaces?**

1. **Dealing with unexpected challenges:** During this experiment, what unexpected challenges or technical issues did you experience with measuring the absorbances of your calibration solutions? What strategy did you use to address this issue, and was it effective?
2. **Considering the absorbance scans you collected, what is the "best" wavelength for measuring caffeine concentration?** Justify your response based on at least two of the following:
 - Beer's Law
 - Sensitivity of the signal to changes in concentration
 - Signal to noise
 - Relative error in the absorption measurement

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4.3: Week 2 - Tea

Pre-lab Assignment

 **Due:** This pre-lab assignment is due prior to the Week 2 Dry Lab meeting.

 **Purpose:** This pre-lab is designed to prepare you for a productive dry-lab discussion and upcoming data analysis. By organizing your results from Week 1 and drafting your Methods section now, you'll be ready to share work-in-progress and get early feedback to improve your report.

Follow up from Week 1

1. **Complete and submit any unfinished business from the last meeting.**
2. **Begin preparing your manuscript document:**
 - a. Complete a draft of your **Materials and Methods** section for **Solution Preparation** and **UV-vis Data Collection**.
 - b. Complete the **Results** section for **solution preparation** and **validation of your solution preparation using UV-Vis**. Include the following elements in your manuscript draft (Results section):
 - i. **Overlaid Spectra Figure**
 - Overlay spectra for all calibration standards.
 - Indicate λ_{max} on the plot with an annotation.
 - Apply formatting standards from previous modules:
 - Descriptive figure title & figure number.
 - Axis labels with units.
 - Caption that describes what is shown and why it matters.
 - Legend, line types, and color scheme that is colorblind-friendly.
 - Use Scan software, MATLAB, or another approved plotting tool.
 - ii. **Calibration Curve Figure**
 - Plot absorbance vs. concentration with best-fit line, equation, and R^2 .
 - Include slope, intercept, and their errors.
 - Apply same figure formatting standards as above.
 - iii. **Calibration Validation**
 - Compare measured value for control solution to expected value.
 - Perform error analysis and state whether your curve is validated.
 - If not validated, use a literature molar extinction coefficient for caffeine to re-determine solution concentrations, and explain discrepancies.
3. **Justifying the Transition to HPLC:** Write a short paragraph for your manuscript explaining why HPLC is the method of choice for final quantitation of caffeine in the beverage samples, even though you have a validated UV-vis calibration.
4. **Use the [Instrument Manual](#) and the [HPLC Simulation](#) described below to prepare for a dry lab discussion about the following:**
 - In an isocratic method, what is the effect of **Solvent B percentage** on peak shape, separation of peaks, and overall retention of the compounds?
 - What is the effect of a **gradient method (from 10%-90% B) compared to an isocratic method at 50% B** on peak shape, separation of peaks, and overall retention of the compounds?
 - What method conditions could you use to isolate caffeine from other compounds and obtain an ideal peak shape? What conditions would not work?

Submit your assignment on Canvas and be prepared to discuss your responses with the team during Dry Lab.

Dry Lab Agenda

In this Dry Lab, you will use the HPLC simulator and group discussion to guide your decision-making about which HPLC method is best for analyzing caffeine in beverage samples.

 **Purpose:** To connect your simulator exploration to real-world method design choices. To compare isocratic and gradient methods in terms of peak shape, separation, and practicality. To prepare a justification for the method you will use in your manuscript.

Instructions

1. **Warm-Up (5 minutes, in groups of two or three):** Share your pre-lab justification with your group. Why do we need HPLC analysis after UV-vis calibration?
2. **Small Group Work (25–30 minutes, in groups of two to three):** Discuss the questions below with your group. Use your simulator observations to support your answers. Record your group's responses in notes, including sketches or comparison tables.
3. **Team Discussion (20 minutes):** Groups will report out. We will compare results and weigh trade-offs between gradient and isocratic methods.
4. **Writing Connection (5 minutes):** Draft 1–2 sentences for your manuscript's Methods section that justify the HPLC method choice.
5. **Start Data Collection!**
 - If you are in the Tuesday Lab section, go to lab now, choose the appropriate method in your sequence template and ask your TA to hit "go" as soon as the samples are at room temperature.
 - If you are in Wednesday or Thursday Lab section, arrange a time to add the method to your sequence file and arrange for your samples to be run overnight.
 - If there is not time to run samples before your lab meeting, you will run them during the lab period this week.

Group Discussion Prompts

Simulator Observations

1. In an isocratic method, how does the percentage of Solvent B affect:
 - Peak shape?
 - Separation of peaks?
 - Overall retention time?
2. In a gradient method (10% → 90% B) compared to an isocratic method at 50% B, how do: Peak shape, Separation, and Retention times differ?
3. Based on your observations, which conditions gave you the clearest caffeine peak?

Method Design Decisions

4. If the best possible separation were your only goal, which method would you choose? Why?
5. If the most practical method for repeated beverage analyses were your priority, which method would you choose? Why? (*Consider this:* between sample injections for gradient methods, the column must be re-equilibrated at the starting %B — this usually *doubles* the time required to run samples using a gradient method!)
6. What are the trade-offs of gradient vs. isocratic methods?

Connecting to Practice

7. Why is a gradient method usually superior for peak shape and separation?
8. Despite that gradient methods are superior in some aspects, why might someone choose an isocratic method?
9. Draft a justification statement for your manuscript that explains why the chosen method is appropriate.

Wet Lab Agenda

Note

There are several UV-vis instruments and several HPLC instruments available to reduce waiting time. In addition, there are some extra computers with the UV-vis software and HPLC software so that students can use the instruments for data collection while other students complete data analysis on a separate computer. Please let your team know if you are waiting to use the instrument or the software.

1. **Repeat your solution preparation as necessary** if your solutions from Week 1 failed to create a valid calibration curve.
2. **Create or finish creating a sequence for running your injections on the HPLC.** Apply the method you chose in the Dry Lab meeting. Your sequence should include injections for the following:
 - Always start a sequence with 2 blank injections.
 - Run *All* beverage samples from your group first if possible (each group member should have a unique type of beverage sample). If all samples are not prepared at the time you create your sequence, communicate with your group members so you can access the sample runs from their data collection. The samples can only be quantitated using your calibration curve if they were run using the same *method* and on the same exact instrument.
 - Another blank to be sure no residue from samples is left on the column.
 - Calibration blank
 - All of your calibration standards — start with the most dilute sample first.
 - Another blank to wash the column after the highest concentration standard.
 - Validation control
3. **Analyze the Sample Data:** Collect the data from at least three unique beverage samples (see above). The instructions for data analysis are in the [Instrument Facilities Manual](#). Be sure to capture the chromatograms of each sample, the peak retention times, peak height, and peak area (from the integration of peaks). It is your choice to collect data for all peaks, or just for caffeine.
4. **Analyze the calibration data:** The instructions for data analysis are in the [Instrument Facilities Manual](#). You should record the retention times, peak height, and peak integration of the caffeine peak in each calibration standard and the validation control. Also figures that would support your results section. The calibration data should be processed using a software of your choice (we do not recommend trying to perform calibration and quantitation with the instrument software).
5. **Continue writing** the *Results* and *Discussion* sections of lab report (only after completing the reflection prompts below.)

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

Reflection Prompt 4.3.0

- **If you were not able to collect HPLC data:**
 - What did you do differently this time when creating calibration standards compared to last time?
 - Did your revised strategy work?
 - What did you learn from additional practice at solution preparation?
- **If you were able to collect HPLC Data:**
 - How did the real data compare to your expectations? Support your response using appropriate evidence.
 - Was the HPLC validation consistent with expectations based on the UV-vis results?
 - You are to include the quantitation of at least three samples in your report (your own sample plus two other unique beverages); How do the measured caffeine amounts compare to literature values.
- **What data will be most useful to support your results section?**

*Reflection prompts should be addressed in your lab notebook and should be included with the lab notebook submission.

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4.4: Week 3 - Tea

Pre-lab Assignment

 **Due:** This pre-lab assignment is due prior to the Week 3 Dry Lab meeting.

 **Purpose:** This pre-lab is designed to prepare you to share work-in-progress and get early feedback to improve your report.

Follow up from Week 2

1. **Complete and submit any unfinished business from the last meeting.**
2. **Prepare your manuscript document:**
 - a. **Update the Materials and Methods section** based on last week's work.
 - b. **Update the Qualitative Results section** based on last week's work.
 - c. **Write your Discussion of Qualitative Results.**
 - d. **Write the Quantitative Results section, including description and figures that show the results of quantitative analysis.**
 - Prepare a table that communicates the results of calibration and quantitation.
 - Prepare one or more figures that shows calibration data.
 - The table and figures should complement each other.
3. Bring a **printed copy** of your manuscript to the Dry Lab meeting.

Dry Lab Group Discussion

In this session, you will peer review each other's reports and practice helping each other by giving impactful, actionable feedback.

Wet Lab Agenda

Record all work in your lab notebook as usual. If you are not doing wet-lab work in this session, please keep still notes of your writing-related conversations in your lab notebook and complete the reflection prompts as usual.

1. **Revise Manuscript and Write Discussion, Conclusions, and Introduction Section.**
2. **Meet with Writing Consultants.**

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

Reflection Prompt **Runaway argument for \eatSpaces?**

1. **How has writing about your lab experience supported your learning?**
2. **How has working with the EWC's supported your writing and learning?**
3. **What have you done that has supported your learning in this module? What could you do better in the future?**
4. **What were the ways in which your instructional team supported your learning? What could they do better in the future?**

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4.6: TA Notes

This page is directed toward laboratory teaching assistants, but is deliberately left open for students to read.

TA Notes

After completion of this module students will be able to:

1. Use analytical techniques to create a serial dilution starting from pure, solid material. (Use an analytical balance, volumetric glassware, and pipettes)
2. Use a full-spectrum UV-vis to measure the absorbance of a series of solutions.
3. Determine the best wavelength to monitor the absorbance of an analyte.
4. Validate a series of standard solutions using a control.
5. Explain the principles of liquid-liquid partitioning.
6. Explain the effect of solvent polarity on retention times.
7. Explain the effects of isocratic and gradient methods on retention times and peak shape.
8. Interpret chromatographic data to identify components of a mixture and determine their concentration.

Pre-Semester Preparations

Do the following during the TA training and orientation week(s)

1. Check that the materials listed below are available and sufficient for all students to use them. If they are running low, please create more or inform the instructor.
 - **Caffeine, 99%:** There should be at least ~25 g of caffeine powder available. If not, please inform the lab director.
 - **Volumetric Glassware:** Be sure that there is sufficient glassware for four students to complete the experiment simultaneously. Please inspect the glassware for chips or obvious contamination. Clean or dispose of glassware as needed and inform the instructor if more glassware should be added to the lab space.
 - **HPLC Autosampler vials:** About 6 vials per student
 - **Syringes and syringe-driven filters:** One each per student
 - Access to UV-vis and at least two quartz cuvettes for each student (Glass cuvettes will not work! Students can take turns.)
2. Complete instrument training with Dr. Haas.
3. Perform the entire experiment as if you were a student.
4. Write and submit a report as if you were a student.
5. Practice grading your and your partner's report using the grading rubric.

Pre-Lab Preparations

When you are on TA duty, arrive at least 20-30 minutes early to prepare and ready the following materials for students.

- Check to be sure that caffeine powder is in its place (notify the instructor if we should order more).
- Check to be sure there are cuvettes available for students to use. They should be **quartz** cuvettes, not glass.
- Remind students to sign into lab
- Check that the student's pre-lab is complete
 - Check their proposed calibration points and ensure that the control solution is approximately in the middle of their range.
- Tell students to gather their glassware as soon as they arrive and get set up on a bench space so that they can make solutions.
- They should clean their glassware prior to use.
- The Cary 100 UV-vis instruments should be turned on at the beginning of lab period. Let the students be involved in this.
- Check that all HPLC modules are powered on (please do not touch the buttons on the bottom components!). The degasser (top unit) is the only one that should possibly be turned off at the button on the instrument.
- Check solvent reservoirs and refill if they are less than half full.
- Show students how to open the software and turn the instrument components on from the software.
- Show students how to start a method running to equilibrate the column to the starting conditions of their first run. Be sure that the pressure is reasonable and the instrument is performing as expected. Otherwise, get help.

Helping your students

During the laboratory session, you are expected to be present and available for the entire 4.5 hours. Your job is to help students to complete the experiment safely and successfully, while upholding expectations and academic standards for the students.

Creating Standard Solutions and Analysis by UV-Vis

In this first laboratory session, please provide explicit instruction on the following:

- **Weighing solid material:** They can begin by weighing their caffeine when their glassware is clean. Please show students how to use the balances correctly and remind them that there is error associated with the measurement. (refer to P-chem materials... we assume that it is in the last digit of the reported number.) Show them how to quantitatively transfer their material into a volumetric flask, rinsing the weighing container into the flask with water.
- **For using volumetric flasks:** Remind them to use good technique. We can assume students know how to use these, but....It takes a while to dissolve the solid material. **DON'T LET THEM MOVE ON TOO QUICKLY!** With powder in flask, fill it about 1/2 to 3/4 with solvent, swirl and sonicate as necessary to dissolve the solid before filling to the line.

Help students to analyze data immediately upon collection. **They should validate their standard curve (UV-vis) using a control solution.** The control solution is provided, and it should be within the concentration range of their calibration curve. If the control is outside their calibration range, a new control should be created. The students should treat the control as a sample in their calibration template, then compare their derived concentration to the known concentration. If the values are the same, then the curve is validated. If not, the students should re-create the standards using correct analytical technique. The students should proceed to the HPLC module only after the calibration is validated.

Encourage students to proceed to the next step and begin creating methods for the HPLC if they have validated their curve and there is additional time in the period.

Analysis of Caffeine by HPLC

Be available to answer questions and troubleshoot any issues as students build HPLC methods and sequence files. Please discuss the following with them:

- General principles of reverse phase chromatography.
- How we predict relative retention of molecules based on their structure and the type of chromatography.
- General expectations for isocratic vs gradient methods.
- Selection of the appropriate wavelength for monitoring caffeine.

During data collection, check to be sure the report is giving peak integrations. If not, please show the students how to manually integrate the peaks and harvest their data electronically.

Remind students to use the control solution to validate their calibration. In this case, if the UV-vis calibration is valid, we expect the HPLC to also be validated. However, if this is not the case, please discuss with the instructor. If the UV-vis calibration was validated, we have confidence that the solutions were made correctly, and there must be another reason that the HPLC calibration has failed.

Shut down/ clean up

15 minutes prior to the end of the period, students should stop lab work and clean up. They are to leave the room in clean condition prior to the end of the period (17:55).

- Remind students when it is time to clean up.
- Remind students to save their samples!
- Turn off the UV-vis
- Remind students that all glassware should be returned to the storage locations.
- Any samples that are stored should be labeled clearly with the student's name and section.
- Remind students to sign out with the time they leave and get your signature.
- Check the TA checklist for additional items to be completed.

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- [Lab 2: High Performance Liquid Chromatography](#) by W. R. Fawcett, John Berg, P. B. Kelley, Carlito B. Lebrilla, Gang-yu Liu, Delmar Larsen, Paul Hrvatin, David Goodin, and Brooke McMahon is licensed [CC BY-NC-SA 4.0](#).

CHAPTER OVERVIEW

5: Tainted Tap! What's in your water?

Learning Objectives

After completing this module students will be able to:

- **Design and justify a calibration strategy** for quantitative elemental analysis, considering when external calibration or the method of standard additions is most appropriate.
- **Analyze and evaluate the influence of matrix effects** on instrument response, and propose strategies to overcome these effects.
- **Apply ICP-OES data to detect and quantify Pb in water**, and assess the results relative to regulatory standards.
- **Interpret differences between data from** unfiltered, tainted, and filtered **samples** to evaluate the effectiveness of a commercial water filter.
- **Integrate scientific literature and regulatory guidelines** to place experimental findings in an environmental and public health context.
- **Communicate research findings** in the format of a scientific manuscript.

[5.1: Introduction - Water](#)

[5.2: Week 1 - Water](#)

[5.3: Week 2 - Water](#)

[5.4: Week 3 - Water](#)

[5.6: TA Notes](#)

This page titled [5: Tainted Tap! What's in your water?](#) is shared under a [CC BY-NC-SA 4.0](#) license and was authored, remixed, and/or curated by [Kathryn Haas](#).

5.1: Introduction - Water

Access to clean, safe drinking water is a critical public health issue. Even at very low concentrations, toxic metals such as lead (Pb) can pose serious health risks, particularly for children.^{1,2} While water in the U.S. is regulated, problems such as the Flint water crisis^{3,4} and other college campus-related discoveries of Pb in drinking water⁵ show that contamination can still occur. Many people turn to consumer water filters as a solution — but how well do they work to remove Pb and other contaminants from water?

In this module, you will investigate these issues by analyzing drinking water samples with inductively coupled plasma–optical emission spectroscopy (ICP-OES). You will design and justify a calibration strategy, assess matrix effects, and evaluate whether a commercial filter removes Pb and other elements effectively. Finally, you will interpret your results in the context of regulatory standards and communicate your findings in the style of a scientific manuscript.

You will work with your group to address these research questions:

- A. Are there detectable amounts of toxic Pb in the water sources across Duke's campus?
 - B. What other metal contaminants are in your drinking water?
 - C. How well does a commercial water filter remove Pb and other metal ions from drinking water?
-

This module was adapted from Gouger, Ashley R., and Jaime E. Mirowsky. “Using Project-Based Learning to Assess the Effectiveness of Water Filtration Devices in Removing Chemical Analytes in an Undergraduate Analytical Chemistry Laboratory.” *Journal of Chemical Education* 99, no. 12 (2022): 4016–23. <https://doi.org/10.1021/acs.jchemed.2c00338>.

References

1. CDC. “About Lead in Drinking Water.” Childhood Lead Poisoning Prevention, August 20, 2025. <https://www.cdc.gov/lead-prevention/prevention/drinking-water.html>.
 2. US EPA, OW. “Basic Information about Lead in Drinking Water.” Overviews and Factsheets. February 2, 2016. <https://www.epa.gov/ground-water-and...drinking-water>.
 3. Wikipedia. “Flint water crisis.” August 10, 2025. <https://en.wikipedia.org/w/index.php...did=1305206269>.
 4. CDC. “CDC Newsroom.” January 1, 2016. https://archive.cdc.gov/www_cdc_gov/...ater-lead.html.
 5. The Daily Tar Heel. “Three Hamilton Hall Drinking Fountains Found to Have Detectable Levels of Lead.” Accessed August 27, 2025. <https://www.dailytarheel.com/article...all-lead-short>.
 6. Wikipedia. “Lead and Copper Rule.” July 16, 2025. <https://en.wikipedia.org/w/index.php...did=1300860169>.
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5.2: Week 1 - Water

Pre-lab Assignment:

 **Due:** This pre-lab assignment is due prior to the Week 1 Dry Lab meeting.

 **Purpose:** Collect background information that can support the introduction section of your report, and prepare to make team decisions about Pb calibration and targets for qualitative analysis. You'll read, calculate, and then propose a plan you can defend to your group.

Pre-Lab Tasks

Read (and add references to Zotero)

- Review the [Methods of Calibration](#).
- Read about how [ICP-OES](#) works, and what you can detect using this method.
- Investigate the EPA's **Lead and Copper Rule**.
 - US EPA, OW. "Lead and Copper Rule." Other Policies and Guidance. October 13, 2015. <https://www.epa.gov/dwreginfo/lead-and-copper-rule>.
 - Wikipedia. "Lead and Copper Rule." July 16, 2025. <https://en.wikipedia.org/w/index.php...did=1300860169>.
- Read [this article](#) about Pb detected in the water on UNC's campus! <https://www.dailytarheel.com/article/2022/09/university-hamilton-hall-lead-short>
- Find at least one reliable source on the health impacts of lead exposure (e.g. CDC, EPA, or peer-reviewed article) and write at least one sentence about the health effects of Pb. Cite your source(s).
- Use reliable sources to identify at least two additional contaminants that could be in your drinking water and that can be detected using ICP-OES. Cite your sources of information.
- Bring your water sample to WET lab (1.0 L of water from a tap at Duke — Due at the start of Wet Lab - collect the sample prior to 1:25 pm)

Pre-Lab Questions

1. According to the EPA:
 - What is the *safe* level of Pb in drinking water?
 - What is the EPA action level for Pb, and how does it shape your expectations for this analysis?
2. Given the EPA action level, what hypotheses do you form about our campus's water, and how will these hypotheses guide your calibration strategy?
3. Design a strategy for preparing the tainted water sample so that it will have 25 ppb Pb. Show your calculations and justify why this concentration is appropriate for the ICP-OES method.
4. Given the choice between **external calibration** and **standard additions**, what factors should your team consider in deciding which to use for this experiment? Describe how matrix effects might influence your decision.
5. Based on your predictions about the water samples and possible matrix effects, which calibration method should your team adopt? Justify your choice by comparing the advantages and limitations of each method in this context. Be prepared to defend your proposal to your team in Dry Lab.
6. If your team adopts the calibration method you proposed, what specific standards will you need to prepare? Show one example calculation and explain why these standards are appropriate for detecting ~25 ppb Pb with ICP-OES.
7. Suppose one team member argues for external calibration because it is simpler and faster, while another argues for standard additions because of possible matrix effects. How would you lead your team to consensus on which method to use?

Water Sample

 **Due:** Collect 1.0 L of water sample prior to the Week 1 WET Lab meeting. Bottles are available for sample collection; they are in the cabinets underneath the ICP-OES in 1112 FFSC.

Each group member should collect one unique water sample. You may take water from unique locations on campus or from the same locations at a different time of day. (See where other groups have collected samples on [google maps](#)). Each sample should be approximately 1 L of water, and collected using a clean metal-free container. The water should be chosen from a drinking water source on Duke's campus.*

Caution

If you are using your own container to collect drinking water for analysis, it must be rinsed with nitric acid to remove residual metals from the inside of the container. Be sure to use a metal-free sample collection vessel.

*If you would prefer to analyze water sources from your home or another place of interest, you are welcome to do so. We can perform qualitative analysis for almost any element - feel free to follow your interests! If you would like to do qualitative analysis on additional samples, they should be collected in smaller 50 mL tubes or in your own containers. You could choose to collect water from a filtered bottle filling station - but some contaminants may be removed already by the filling station filter.

Dry-Lab Group Discussion:

In this discussion, you will practice making scientific decisions as a team. Instead of simply following a procedure, you will weigh evidence, anticipate challenges such as matrix effects, and reach consensus on an analysis plan. By the end of this session, your group will have chosen a calibration strategy and identified additional elements for qualitative analysis. These decisions will prepare you to enter Wet Lab with a clear plan, so that your calibration standards and water samples can be prepared efficiently and consistently. This work also develops transferable skills in collaborative problem-solving, justifying methodological choices, and documenting group decisions—skills that are essential for authentic scientific research.

 **Goals:** By the end of today's discussion, your team will:

- Choose one calibration strategy (all members use the same).
- Finalize a Pb calibration set (levels, glassware plan, who prepares what).
- Select ≥ 4 qualitative targets beyond Pb and record your rationale (how they help interpret filter performance and matrix effects).

Instructions:

- Begin the session by grouping up and assigning roles (scribe, reporter, spokesperson, time keeper)
 - The Time Keeper will start a timer and encourage group consensus.
 - The Scribe will record the group work.
 - The Spokesperson will summarize your group's decisions to the class in 2 minutes at the end of discussion.
 - The reporter will submit the group record at the end of class.
- Work on the discussion questions together in your group for ~25 minutes.
- Report out
 - What are the elements chosen for qualitative analysis?
 - Describe your calibration strategy and justify this choice.
- As a group, make a detailed plan for creating calibration standards so that you can begin solution preparations immediately at the start of wet lab.

Group Discussion Prompts

1. Elements for Qualitative Analysis

- Each group member brings at least two candidate elements from the pre-lab. As a group, discuss:
 - Which elements are most likely to be present in drinking water at detectable levels?
 - **Decision:** Select at least three elements for qualitative analysis (beyond Pb).
 - **Action:** Record your group's list with a short justification for each.

2. In this module, we are addressing the question "How well does a water filter work?". Considering this, **explain why we are creating a tainted water sample** instead of just working with the plain drinking water sample.

3. Calibration Strategy for Quantitative Analysis of Pb

- Share your individual pre-lab calibration proposals. What evidence or reasoning did each of you use?
- If you use external calibration, how can you validate your calibration and determine whether there are matrix effects?
- In the context of this project, what are the pro's and con's of calibrating with the method of standard additions instead of external calibration?
- **Decision Point:** Which calibration method will your group use?
- **Action:** Write down your choice and the main reasons (bullet points are fine).

4. Protocol for Standard Preparation: Review your pre-lab calculations for tainting and standard preparation.

- Is the range of standards appropriate for measuring ~25 ppb Pb? Explain your reasoning.
- **Decision Point:** Finalize the set of calibration standards that your group will prepare in Wet Lab. Your entire group should agree on the range and concentrations of each standard. Each group member will prepare their own calibration replicate(s).
- **Action:** Write down the agreed concentrations and required glassware/pipettes. Create a group protocol for preparing the standards.

5. In the context of analyzing drinking water with ICP-OES, what kinds of questions can we answer using qualitative analysis, and what questions require quantitative analysis? Give at least one example for each from your group's water project.

6. How might the choices your group made today appear in the Introduction or Methods section of your manuscript? Draft one sentence that could appear in your Methods section to describe your calibration strategy.

Wet-Lab Work:

 **Goals for Wet Lab:** By the end of Week 1 Wet Lab, aim to accomplish:

- Prepare three samples for QUAL analysis (~10-15 mL each).
- Build a group method file for QUAL analysis.
- Run all samples using the QUAL method and harvest data in a form that is useful for your report.
- Make progress toward calibration of the instrument for QUANT analysis.

Agenda for Wet Lab

1. **Prepare containers** (remove trace metals)
2. **Prepare samples**
 - source (untainted) samples
 - tainted water samples
 - filtered water samples
3. **Collect qualitative data on the samples.**
4. **Prepare calibration standards and spiked recovery samples**
5. **Collect calibration and sample data** to determine the Pb concentration (if time allows)
 - Create a calibration curve
 - Use your calibration to determine the amount of Pb in your water samples.
 - Validate your method by performing a spiked recovery if applicable - If you cannot validate your method, revise your strategy and try again.
6. **Store all samples** in sealed containers and clean up.

Instructions for Completing Wet Lab Work

1. Prepare your containers

Remove trace metals from any glassware or reusable sample containers!

All sample containers and glassware must be rinsed well with dilute nitric acid (0.1 M) to dissolve and remove any trace metal ions that are bound to the interior surface of the container. Expose the interior of the container to 0.1 M nitric acid (HNO₃) in deionized water for 10 minutes, then rinse with deionized water.

2. Sample Preparation

Each group member will prepare three types of samples for this analysis using their unique source water. Prepare each sample for analysis by filling an autosampler tube about 4/5 full with the sample:

1. **Source water sample:** This is the water directly from the source tap (aka "untainted water"). Reserve approximately 100 mL of your source water sample for direct analysis using the ICP-OES. In other words, save some and don't taint it or filter it so we can measure its "original" contents. Fill an autosampler tube with about 10-12 mL of this water for QUAL analysis. You will use the remaining ~90 mL for your QUANT calibration standards or validation.
2. **Tainted water sample:** We *really* hope, and expect, that you won't find much, if any, toxic metal in the water on our campus! So...to give you some Pb to work with, you will use a certified element standard to contaminate **500 mL of water** sample to a final concentration of **25.0 ppb Pb**. This is your "tainted" water sample. After you prepare the tainted water solution, prepare a sample for analysis by putting about 10-12 mL in a labeled 15 mL tube. Reserve some of this tainted water for calibration or validation.
 - o The **detection limit of ICP-OES is close to 5 ppb for Pb** under some conditions. Adding a known amount of Pb to your sample will give you a better chance to measure any trace amounts less than the limit of detection that are in the water sample.
3. **Filtered water:** You will use a Brita water filter cartridge to filter a portion of your tainted water and then measure the Pb concentration of the filtered water.
 - o **Steps to prepare the filtered water:**
 - a. Start by rinsing the filter and pitcher with deionized water three times.
 - b. Remove all water from the pitcher and gently shake the pitcher to remove excess water from the entire filter unit.
 - c. Prepare the filter: Filter a small portion of your tainted water three times, discard the filtered water each time into a waste container.
 - d. Filter a fourth portion of the tainted water, and collect the filtrate into a clean container. Collect enough filtered sample for the Qual analysis, as well as for calibration or validation. Transfer 10-12 mL of the filtered water to an autosampler tube. This is the filtered water sample for analysis.

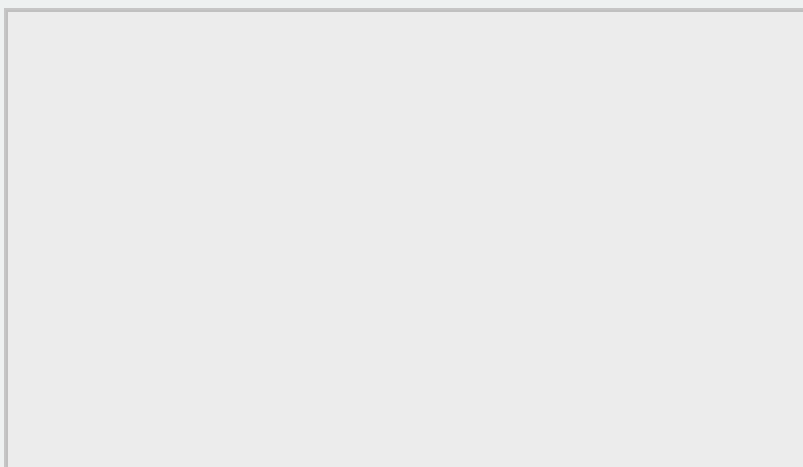


Figure 5.2.1: Scheme for sample preparation (CC BY SA; Kathryn Haas)

✓ 3. Qualitative Analysis of samples (what's in this water?)

As a group, create an ICP-OES instrument method and analyze all of your group's samples (3 samples per person). Your qualitative analysis method should include **Pb** and all the **elements that your team chose** for analysis (from your Dry Lab decisions).

- Instructions for **creating a method** for qualitative analysis are in the [Instrument Facilities Manual \(click here\)](#).
- Instructions for **operating the instrument** are in the [Instrument Facilities Manual \(click here\)](#).
- Instructions for **viewing and exporting qualitative data** are in the [Instrument Facilities Manual \(click here\)](#).

✓ Your Data Checklist

To prepare for analysis and manuscript writing, be sure to export and save the following items to your own computer or cloud storage before you leave the lab:

- **For each element and each water source:** Overlaid emission spectra of that element in the blank, in the source water, the tainted water, and the filtered water. (If there were two unique water sources and three elements, this would be six separate figure panels).
- **Export the emission peak values** (the results table) for all samples and all elements, including the blank

📌 4. Prepare Calibration Standards

The calibration method is up to you! Review the different methods of calibration ([single vs multi-point](#), [external calibration](#), [standard addition](#)) that you have or will learn about in the complimentary Analytical Chemistry Lecture Course. You should choose a method of multi-point calibration and plan how you will create the standard solutions necessary to calibrate the instrument. The goal is to accurately measure a solution of approximately 25 ppb Pb. Your group will be provided with the following materials:

- Up to **250 mL** of a certified Pb standard that contains 1.000 ± 0.002 mg/L (ppm) Pb.
- deionized water
- 0.1 M nitric acid in deionized water
- any flasks and containers available in the lab
- digital pipettes (for delivering volumes less than 1 mL, or odd volumes <5 mL for which volumetric pipets are unavailable)
- volumetric pipets (for most volumes > 1 mL)

You will be given a certified Pb standard that you can use to create the standard solutions or spikes necessary to calibrate the method. You should use volumetric glassware and techniques as you've practiced earlier in this course. If you need to transfer a volume of <5 mL that does not correspond to a typical capacity of a volumetric pipet, a digital pipette can be used to transfer volumes between 10-5000 μL . Take note of the errors associated with different types and sizes of pipet(te)s (see orientation module). Make sure that all the glassware is cleaned well to remove any trace contamination (see instructions above).

✓ 5. Quantitative Analysis of Pb (how much?)

As a group, create an ICP-OES instrument method and analyze all of your group's calibration standards and samples. Your quantitative analysis method should include only **Pb** for elemental analysis and calibration.

- Instructions for [creating a method for quantitative analysis](#) are in the Instrument Facilities Manual ([click here](#)).
- Instructions for [operating the instrument](#) are in the Instrument Facilities Manual ([click here](#)).
- Instructions for [viewing and exporting quantitative data](#) are in the Instrument Facilities Manual ([click here](#)).

The analysis of all samples will take a while, and can be automated to run while you are away. While these samples are running, work on harvesting qualitative data.

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

📌 Reflection Prompt 5.2.0

We assume you haven't gotten a chance to fully collect or analyze calibration data at this point, so these questions are focused on *qualitative* data.

1. **Qualitative Analysis Results:** What elements did you detect in your source water, tainted water, and filtered water? Which of these findings were expected based on your pre-lab predictions, and which were surprising?
2. **Matrix Effects and Calibration Choices:** Based on your results, what evidence do you see of possible matrix effects? How would you evaluate whether your calibration strategy (external vs. standard additions) is an effective choice?
3. **Filter Effectiveness:** How did filtration affect the signals for Pb and your team's additional elements? Did the filter perform as you expected? What evidence supports your conclusion?
4. **Reproducibility:** How reproducible are your results based on replicate measurements or repeated scans? If variation differs by signal intensity, what might explain that trend?
5. **Data Presentation & Communication:** Include one overlaid spectrum (at least one analyte across source/tainted/filtered).
 - Provide a short figure caption that communicates what the spectrum shows and why it matters.
 - Write 2–3 sentences explaining how you might present these results in the Results section of a scientific manuscript.

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5.3: Week 2 - Water

Pre-lab Assignment

 **Due:** This pre-lab assignment is due prior to the Week 2 Dry Lab meeting.

 **Purpose:** This pre-lab is designed to prepare you for a productive dry-lab discussion and to prime you to reasonably interpret your data. By organizing your qualitative results and drafting your Methods section now, you'll be ready to share work-in-progress and get early feedback to improve your report.

Follow up from Week 1

1. **Complete and submit any unfinished business from the last meeting.**
2. **Begin preparing your manuscript document:**
 - a. **Complete a draft of your Materials and Methods section.**
 - b. **Complete the Qualitative Results section, including description and figures that show the results of qualitative analysis.**
 - Prepare a table of the elements you searched for in qualitative analysis for all samples, and that indicates their presence in your sample.
 - Prepare a figure or multiple figures that shows the results of qualitative analysis.
 - The table and figures should complement each other.
 - c. **Draft your Discussion of qualitative results:** Analyze and interpret the qualitative results for all the samples from your group:
 - **For all source water samples:** Explain whether each element is present or absent in detectable amounts in each sample, using evidence from your results section to justify your analysis.
 - **Compare source water to tainted:** Does the addition of Pb to the matrix (tainting) affect the signals of other elements? Support your assertions using evidence from your results section. Explain the meaning of these results in the context of matrix results.
 - **Compare tainted sample to filtered sample:** What effect does filtration have on the intensity of the signals from each element? Support your assertions using evidence from your results section. Explain the meaning of these results in the context of matrix effects.

Many people assume that signal intensity reflects the amount of an element present, but this is not always true! Matrix effects can make signals appear stronger or weaker without a real change in concentration. In this pre-lab, you will read about Brita's claims for filtration, make predictions about how well the filter should work for Pb and other elements, and consider why signals might increase after filtration. This work will prepare you to identify when a signal change is likely due to matrix effects versus true contamination, and to defend your interpretation during group discussion.

Pre-Lab Questions

Read the claims about the efficiency of Brita water filters for removal of substances from drinking water. (<https://www.brita.com/why-brita/better-water/>). We use a *standard* Brita pitcher filter in this experiment.

1. **Compare Predictions to Evidence:** After reading Brita's claims, what efficiency do you expect for Pb removal? What efficiency do you expect for your group's additional elements? State your predictions clearly so you can compare them to your qualitative results now, and your quantitative results later this week.
2. **Signals that Increase After Filtration:** Sometimes, the intensity of an element's signal *increases* after filtration. What are two possible explanations for this observation — one related to matrix effects and one related to an actual change in sample composition? Which explanation would you expect to be more common, and why?
3. **Responding to Unexpected Results:** Suppose your group observes higher intensity signal for an element in the filtered sample than in the tainted sample. How would you decide whether this is a real contamination event (e.g., improper filter preparation) versus a matrix effect? What additional evidence or checks could help you distinguish between these possibilities?

Dry Lab Group Discussion

 **Goals:** By the end of today's discussion, your Team will:

- Compare qualitative results across all Groups to look for evidence of matrix effects.
- Identify which elements increased or decreased after filtration, and debate possible explanations.
- Distinguish between matrix effects and true contamination events using shared evidence.
- Draft clear sentences for the Results and Discussion sections of your manuscript that reflect both Group and Team-level evidence.

Instructions

1. Step 1 — Group Check-In (10 minutes)

Within your 4-person Group:

- a. Compare your pre-lab predictions to your Group's actual qualitative results.
- b. Decide: Did we observe any element with an increased signal after filtration?
- c. Nominate one spokesperson to report back to the Team.

2. Step 2 — Team Data Pooling (15 minutes)

Each Group's spokesperson shares findings with the Team (3 short reports).

- a. As a Team, nominate two scribes; one to write on the whiteboard, and another to record in a shared google sheet.
- b. Create a combined table (on the whiteboard and in the shared document)
 - Columns = Elements tested
 - Rows = Group A, Group B, Group C
 - Record whether signals after filtration were ↑ (increased), ↓ (decreased), or = (no change).

3. Step 3 – Team Interpretation (15 minutes)

As a Team, address the following:

- a. Which elements showed signal increases in at least one Group's data?
- b. Are these more consistent with matrix effects or true contamination? What evidence supports your conclusion?
- c. If Groups disagree, what additional checks or information could help resolve the discrepancy?

4. Step 4 – Group Drafting (20 minutes)

- a. Re-group with the members of your section and set a timer for 10 minutes: Each person in the group should spend 10 minutes revising the Discussion draft that they wrote for their pre-lab assignment. Focus on
 - At least one results sentence describing a shared observation (e.g., "In all filtered samples, signals for Cu decreased relative to tainted water.").
 - At least one discussion sentence interpreting it (e.g., "This decrease is consistent with decrease in Cu contamination in the sample as a result of filtration.").
- b. Take turns reading your revised sentences aloud, and giving feedback to each other. Spend the remaining time in dry lab to refine your Discussion section — even though you are each writing your own manuscript, your group should reach consensus on the interpretation of your qualitative evidence as a whole.

Wet Lab Agenda

If you were not able to initiate quantitative data collection in week 1, please prioritize the solution preparation and data collection. Then proceed with the steps below.

1. **Access your data:** You will need to know your file names and in which directory you stored your data so you can locate the data and complete the analysis.
2. **Perform Quantitative Analysis:** Depending on how you calibrated your samples, you will work with your group or individually to process and export the quantitative data.
 - Produce a calibration curve using the appropriate regression model.
 - Assess the validity of your calibration.
 - Perform quantitative analysis on your samples.

- *Make sure to include error analysis!*

3. **Continue writing** the *Results* and *Discussion* sections of lab report (only after completing the reflection prompts below.)

✓ Your Data Checklist

To prepare for analysis and manuscript writing, be sure to export and save the following items to your own computer or cloud storage before you leave the lab:

Calibration Standards Data:

- Overlaid emission spectra for all Pb calibration standards.
- Pb emission peak values for each standard exported in a format that you can process using your favorite software.
- A calibration curve for Pb (plotted using your favorite software), including best-fit line and regression statistics
- Documentation of your calibration strategy (external vs. standard additions) and its validation (e.g., spiked recovery).

Sample Data:

- Overlaid emission spectra of Pb in all water samples (source, tainted, filtered) run at the same time as your calibration data.
- Pb emission peak values for each sample
- A table of Pb concentrations in each water sample, with error estimates.

Evidence for Notebook Snapshot

- Any exported files or screen captures that show results, with notes to show your understanding of their relevance and meaning.

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

Reflection Prompt 5.3.0

In your notebook submission, respond to the following:

1. **Results:** Briefly summarize what you understand about your quantitative results.
2. **Filter Effectiveness:** Based on your quantitative results, how effective was the filter at removing Pb from tainted water? Do you know if the filter remove other elements? What evidence supports your conclusion?
3. **Validation:** Did your calibration strategy pass validation? If not, what does this mean for your interpretation of filter effectiveness, and what changes must you make in your calibration approach before repeating the analysis?

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5.4: Week 3 - Water

Pre-lab Assignment

 **Due:** This pre-lab assignment is due prior to the Week 3 Dry Lab meeting.

 **Purpose:** This pre-lab is designed to prepare you to share work-in-progress and get early feedback to improve your report.

Follow up from Week 2

1. **Complete and submit any unfinished business from the last meeting.**
2. **Prepare your manuscript document:**
 - a. **Update the Materials and Methods section** based on last week's peer feedback.
 - b. **Update the Qualitative Results section** based on last week's peer feedback.
 - c. **Write your Discussion of Qualitative Results.**
 - d. **Write the Quantitative Results section, including description and figures that show the results of quantitative analysis.**
 - Prepare a table that communicates the results of calibration and quantitation.
 - Prepare one or more figures that shows calibration data.
 - The table and figures should complement each other.
3. Bring a **printed copy** of your manuscript to the Dry Lab meeting.

Dry Lab Group Discussion

In this session, you will peer review each other's reports and practice helping each other by giving impactful, actionable feedback.

Wet Lab Agenda

Record all work in your lab notebook as usual. If you are not doing wet-lab work in this session, please keep still notes of your writing-related conversations in your lab notebook and complete the reflection prompts as usual.

1. **Revise Manuscript and Write Discussion, Conclusions, and Introduction Section.**
2. **Meet with Writing Consultants.**

Notebook

Record detailed notes about your lab work in your notebook each time you are in lab. Also include the data analysis and responses to the reflection prompts*. Your notebook should be submitted on canvas prior to leaving Wet Lab each week - whatever state it is in at the end of lab is the state in which you should submit your work. If you are unable to complete data analysis and the reflection prompt within the period, it should be completed before the next lab period, and submitted with your next notebook submission.

Reflection Prompt **Runaway argument for \eatSpaces?**

1. **How has writing about your lab experience supported your learning?**
2. **How has working with the EWC's supported your writing and learning?**
3. **What have you done that has supported your learning in this module? What could you do better in the future?**
4. **What were the ways in which your instructional team supported your learning? What could they do better in the future?**

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5.6: TA Notes

This page is directed toward laboratory teaching assistants, but is deliberately left open for students to read.

TA Notes

After completing this module your students will be able to...

- describe external calibration and the method of standard additions.
- describe the pros and cons to using different calibration methods.
- describe how to validate a calibration using a spike.
- explain what a matrix effect is, and how to recognize it.

Pre-Semester Preparations

Do the following during the TA training and orientation week(s)

1. Check that the materials listed below are available and sufficient for all students to use them. If they are running low, please create more or inform the instructor.

- **0.1 M Nitric Acid** should be prepared by dissolving 66 mL of 70% nitric acid (15.8 M) in 10 L of ultrapure water.

Caution

This should only be done by a chemist trained to safely handle concentrated nitric acid. 70% nitric acid penetrates standard lab gloves and acid-resistant 5 mil neoprene gloves should be worn.

- **Pb standard solution, 1000 ppm is provided.** This solution is much too concentrated for the students to use directly in their calibration. Thus, a more dilute solution should be made first. A solution of 1 ppm = 1000 ppb is more suitable as a working solution.

Caution

Pb is very toxic to humans and other living organisms. Wear PPE and handle carefully.

- **Calibration solutions:** A five-point calibration (including the blank) is typical, and a calibration curve should range between 0 and 2-3 times the concentration of the analyte in the samples. TAs should carefully prepare excess external calibration solutions as they complete the module, and have them on hand for instrument performance checks during the semester.
 - 1g/L Na solution should also be prepared or available to run a sodium bullet test.
2. Complete instrument training with Dr. Haas.
 3. Perform the entire experiment as if you were a student.
 4. Read this: [Guided-Inquiry Experiment for Teaching the Calibration Method of Standard Addition in the Analysis of Lead with Graphite Furnace Atomic Absorption Spectroscopy | Journal of Chemical Education \(acs.org\)](#)
 5. Write and submit a report as if you were a student.
 6. Practice grading your and your partner's report using the grading rubric.

Pre-Lab Preparations

When you are on TA duty, arrive at least 20-30 minutes early to prepare and ready the following materials for students.

- **0.1 M nitric acid:** this should already be prepared using metal-free nitric acid by the lab director, and excess solution should be available in a large carboy in the hood or prep room. As students arrive to lab, remind them that they should rinse their glassware with this solution to remove all residual metal from the glass.
- **1 L of drinking water** collected by students: As your students arrive on the first day of the module, check that they have collected water or ask them to do it immediately.

- **Pb standard solution, 1000 ppm:** This is a solution purchased and used as received. If it is running low please inform the instructor.
- **Deionized water:** Please check that the ultrapure water is running and there are no leaks in the system under the sink.
- **Brita filter:** Please check that the Brita filter is available and begin rinsing the filter with ultrapure water in preparation for the students to use it.
- Remind students to sign into lab
- Check that the student's pre-lab is complete

Helping your students

During the laboratory session, you are expected to be present and available for the entire 4.5 hours. Your job is to help students successfully complete the experiment while upholding expectations and academic standards for the students.

In this particular laboratory session, you should let students choose their calibration strategy. You can discuss the pros and cons of different strategies, but it should ultimately be the decision of the student group as to what method they choose to calibrate the instrument. You should also discuss validation with them, but it is their choice as to how they validate their method or whether they do the validation at all.

This lab module is meant to be open-ended and also to demonstrate matrix effects. Please discuss matrix effects with the students and make them aware of this as a possible explanation of their results. The students can, and should, use lab time to test any hypotheses they develop to explain their results. For example, if the signal changes after filtration, and the students think there is a matrix effect, that hypothesis can be tested by the method of standard additions or by a simple spike to each of the samples in question. Alternatively, if students believe the filter is adding metals to their filtered sample, that hypothesis can be tested by measuring the metal signal of ultrapure water before and after exposure to the filter.

Encourage students to use *at least* two full lab periods (assuming a 3-period module) to perform lab work. Ideally, data is actively processed during collection and students are able to finalize data collection and begin writing their reports during the third week of lab.

Shut down/ clean up

15 minutes prior to the end of the period, students should stop lab work and clean up. They are to leave the room in clean condition prior to the end of the period (17:55).

- Remind students when it is time to clean up.
- Remind students that all glassware should be returned to the storage locations.
- Any samples that are stored should be labeled clearly with the student's name and section.
- Remind students to sign out with the time they leave and get your signature.
- Check the TA checklist for additional items to be completed.

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